

Foundation Models for Remote Sensing Image

Processing: Architectures, Pre-training Strategies, and

Modal Adaptation

Abstract:

The evolution of remote sensing into a data-rich domain has been made possible by the rapid growth of available satellite images with varying types of resolutions (spatial, spectral, or temporal). Traditional task-specific types of models tend to have difficulty generalizing across different types of sensors, geospatial areas, and various applications, thus showing the need for scalable and adaptable types of learning paradigms. Foundation models represent a novel way of addressing this issue and provide a transformative mechanism to take advantage of large amounts of self-supervised data collected from a wide area over time (e.g., Earth observation). This paper provides an extensive review of foundation model implementations for processing remote sensing images. The emphasis will be placed on both the evolution of architecture (CNN and Vision Transformer) and types of pre-training (e.g., Masked Image Modeling, Contrastive Learning, Knowledge Distillation). The analysis will show how Vision Transformers can model global spatial connectivity much better than CNNs as well as how they can support the integration of multi-modal data (e.g., imagery and metadata). Moreover, we will discuss the pros and cons of each of the three main types of pre-training methodologies with respect to inductive bias and appropriateness to each specific task. In addition, we describe current models based on their abilities to meet spatial, spectral, and temporal constraint needs while reviewing their performance on various downstream applications like classification, segmentation, and temporal prediction. We also address how large data sets help support scalable learning. Finally, we identify challenges and offer future directions for developing physical, general-purpose remote sensing foundational models.

Keywords: *Remote Sensing, Foundation Models, Vision Transformers, Self-Supervised Learning, Masked Image Modeling, Multimodal Remote Sensing*

1. INTRODUCTION

For many years, remote sensing, as a field, has been the primary mechanism for monitoring the Earth's dynamic systems, providing important insights on climate change, urban planning, disaster response, and precision agriculture. Remote sensing has long served as a primary tool for obtaining information about an object, area, or phenomenon without making physical contact with it [1]. The rapid evolution of remote sensing applications has significantly altered its application methods. Previously, task-specific workflows dominated the field, enabling the execution of various downstream tasks. Remote sensing worked by collecting a limited set of data and then manually labeling them before training supervised algorithms like random forests and support vector machines (SVM), later transitioning into convolutional neural networks (CNN) [2]. These methods were effective for singular tasks for a smaller range of applications. Unlike natural images, labeling this satellite imagery with complex spectral signatures created challenges, as it required domain expertise, making the process timely and cost-inefficient.

Simultaneously, the observed data has increased rapidly due to the daily revisits from sources like Sentinel, Landsat, and commercial providers. While we possess abundant raw data, traditional supervised models cannot process it effectively. The approach lacked generalization, meaning for every new region, sensor, or task, we needed to train a new model from scratch, limiting the scalability of remote sensing applications. Accurate and efficient usage of remote sensing data remained a challenge.

To overcome these challenges, foundational models are being adopted. "A foundation model is any model that is trained on broad data (generally using self-supervision at scale) that can be adapted (e.g., fine-tuned) to a wide range of downstream tasks" [3]. Remote sensing foundational models use a "pre-train and fine-tune" approach. Rather than being trained on a specific small labeled dataset, foundational models are pre-trained on huge amounts of unlabeled satellite imagery using Self-Supervised Learning (SSL) techniques, such as Masked Autoencoders (MAE) [4]. MAE trains the model to reconstruct a missing part of an image, helping it learn more robustly about the intrinsic features of the earth's surface.

Foundational models are important because they provide a general backing for various remote sensing applications. These models have a generalized understanding of the Earth, allowing it to be molded for any other task. Several such models have emerged in the last few years, some were pre-trained for a general use case such as Prithvi-EO 2.0 [5], while others were trained on a specific application such as CloudVit [6].

This paper reviews the current state of foundational models for remote sensing across various types of FMs. It elaborates on the Vision Transformer (ViT) [7] architecture, along with the pre-training strategies. We explored the field from different perspectives, from the background of remote sensing and the historical evolution of RS analysis to the downstream applications of such models and their performance. These sections, along with others, are thoroughly reviewed in this report.

2. REMOTE SENSING DATA CHARACTERISTICS AND EVOLUTION

Remote sensing has become a fundamental technology for large-scale Earth observation, enabling continuous monitoring of environmental dynamics, urban expansion, and climate-driven changes [1]. Data acquired from satellite, airborne, and near-surface platforms provide rich geospatial information across multiple spatial, spectral, and temporal dimensions. The tasks involved in remote sensing images include the acquisition, processing, and analysis of data for understanding the Earth's surface. The major downstream tasks include image classification, object detection, segmentation, image restoration, and image fusion. To design effective foundation models for remote sensing, it is essential to understand the intrinsic properties and complexities of geospatial data distributions.

Remote sensing data is divided into several different types of resolution, including spatial, spectral, temporal, and radiometric [8]. These different types of resolution affect how foundation models represent and process geospatial data. For example, the spatial resolution of the data will determine how detailed objects are represented; the spectral resolution will determine how well materials can be distinguished from one another; and the temporal resolution will determine how

dynamic and evolving, an environmental process can be modeled. Naturally occurring images are typically only represented using either one or two dimensions; therefore, these different forms of resolution create challenges for models to be aware of the different forms of resolution, to be able to accommodate multiple scales of data, and to be able to accommodate temporal information at different rates of change.

Remote sensing data is inherently multi-modal, encompassing optical imagery, synthetic aperture radar (SAR), LiDAR, multispectral, hyperspectral, and thermal infrared observations. Each modality captures distinct physical properties of the Earth's surface, resulting in complementary spatial, spectral, and temporal information. This heterogeneity introduces significant challenges for representation learning, as models must effectively integrate diverse data distributions and sensing mechanisms. As illustrated in Fig. 1, each sensing modality captures distinct physical properties of the Earth, necessitating foundation models that can learn unified representations across heterogeneous data sources.

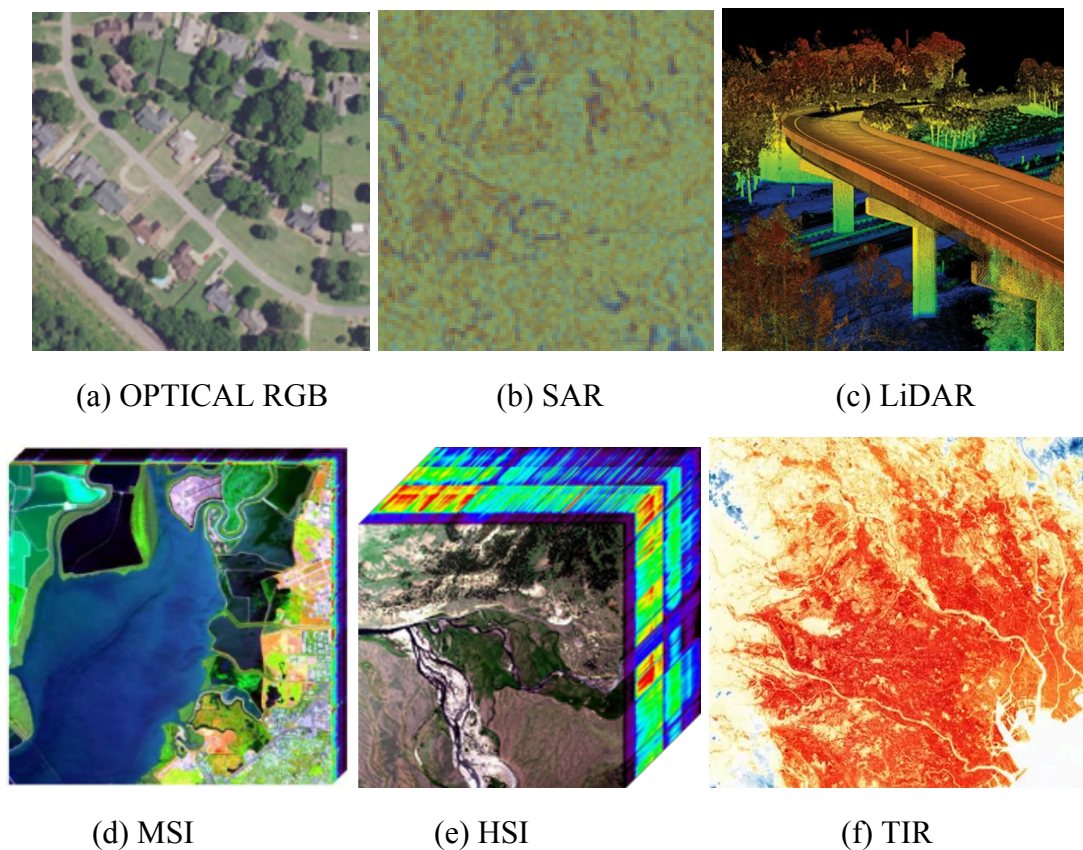


Fig. 1. Examples of different modalities: (a) OPTICAL RGB (Source: [48]), (b) SAR (Source: [59]), (c) LiDAR (Source: [98]), (d) MSI (Source: [72]), (e) HSI (Source: [57]), (f) TIR (Source: [73])

2.1 HISTORICAL EVOLUTION OF REMOTE SENSING ANALYSIS

Remote sensing image analysis has evolved through several distinct paradigms, driven by advances in sensing technology, data availability, and computational methods. Early approaches relied heavily on human interpretation, while recent developments emphasize scalable, data-driven representation learning. The historical evolution of remote sensing image analysis is illustrated in Fig. 2, highlighting the transition from manual interpretation to scalable foundation models. As shown in Fig. 2, each phase reflects a shift in data availability, computational capability, and model design, leading to increasingly automated and generalizable approaches.

The first phase, before the 1970s, was dominated by visual interpretation of aerial photographs. Analysts manually examined grayscale imagery using visual cues such as tone, texture, shape, shadow, and pattern, often aided by stereoscopic viewing to infer three-dimensional structure. While effective for small-scale analysis, this process was labor-intensive, subjective, and incapable of exploiting non-visible spectral information, severely limiting scalability and consistency.

The second phase, spanning the 1990s to early 2010s, emerged with the widespread availability of digital satellite imagery from platforms such as Landsat. This era introduced traditional machine learning methods, where manually engineered features were fed into statistical classifiers such as support vector machines, random forests, and maximum likelihood estimators. Although these approaches improved automation, their performance remained constrained by the quality of handcrafted features and struggled to capture complex, non-linear patterns in high-dimensional remote sensing data.

The third phase, beginning around 2012, marked the adoption of deep learning, particularly convolutional neural networks (CNNs). CNNs enabled end-to-end feature learning and significantly advanced tasks such as land cover mapping and semantic segmentation through architectures like U-Net and ResNet. However, their reliance on large volumes of annotated data

and limited ability to model long-range spatial relationships across large satellite scenes posed critical scalability challenges [1].

The current phase, starting in the early 2020s, is defined by the rise of foundation models trained using self-supervised learning on massive archives of unlabeled satellite imagery. Leveraging architectures such as Vision Transformers and pretraining objectives like masked image modeling, this paradigm aims to learn general-purpose representations that can be adapted efficiently to diverse downstream tasks. By decoupling representation learning from task-specific annotation, foundation models address key limitations of prior approaches and represent a fundamental shift toward scalable remote sensing image analysis.

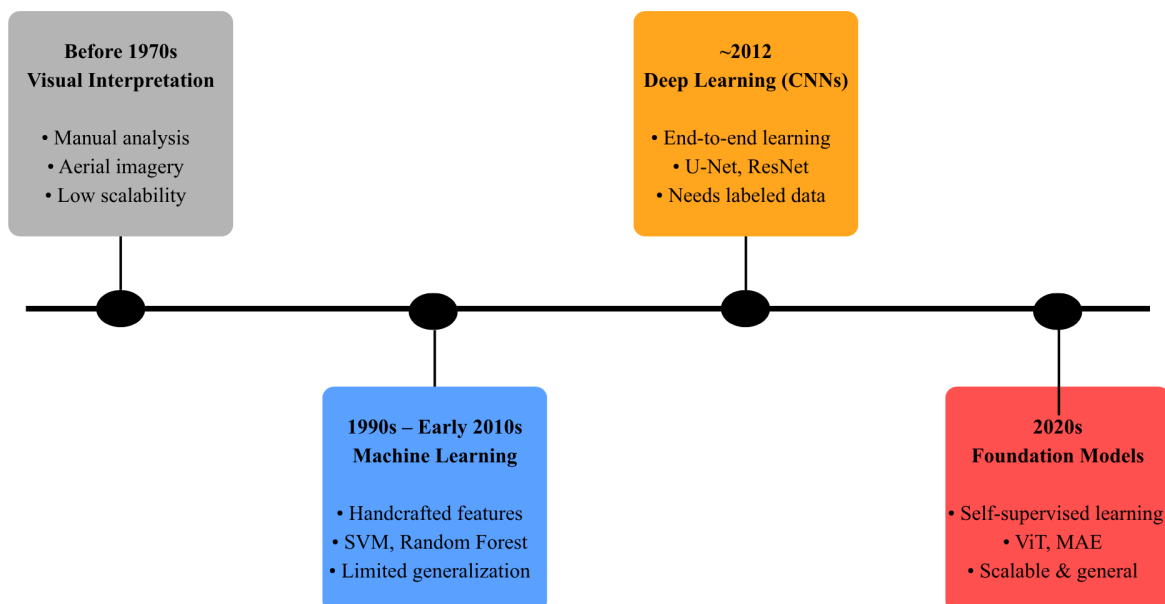


Fig. 2. Historical evolution of remote sensing image analysis. The field has progressed from manual visual interpretation to traditional machine learning, deep learning, and modern foundation models. Each phase reflects advancements in data availability, computational power, and representation learning.

3. ARCHITECTURAL FOUNDATIONS IN RS

The architectural structure of a foundational model greatly influences the representation and transfer of spatial, spectral, and temporal data in relation to different tasks. When it comes to

remote sensing, architectural choices will affect how well a model is capable of generalizing over multiple resolutions, types of sensors, and different geographical areas. In this section, we provide a review of the evolution of architectural paradigms in remote sensing, detailing why convolutional neural networks were unable to serve as adequate foundation backbones and how Vision Transformers emerged as the primary alternative.

The development of foundational models in remote sensing represents a fundamental architecture shift from earlier Convolutional Neural Networks (CNNs) [12] dominated deep learning. The current state-of-the-art models are built on Vision Transformers (ViT) [7]. This evolution is defined by a transition from architectures that prioritize local features (Convolutional Neural Networks) to those that prioritize global context (Vision Transformers). Understanding the architectural foundations is essential to understanding foundational models.

3.1 The Predecessor: Convolutional Neural Networks (CNNs)

In the past ten years, the success of Convolutional Neural Networks (CNNs) as a new and major paradigm in remote sensing image analysis has come from their architectural inductive biases toward local spatial continuity and translation equivariance within the input space [12], which strongly corresponds with Earth's texture-dominated EOS data, where classes of objects, such as vegetation, built environment, and water bodies, are characterized by local repeated patterns and not by global geometries [12].

The Architectural Advantages of CNN's Inductive Bias:

Even though they were not developed specifically to process remote sensing imagery (e.g., satellite images), CNN's architecture has hard-coded assumptions that hold true when using satellite image data. By utilizing the convolution operation and applying weight sharing, a CNN can efficiently learn local spatial hierarchies.

Texture Capture - CNNs are particularly good at capturing high-frequency, lower-level local textures that are important for classifying land cover; for example, SAR backscatter patterns or crop row images [2].

Translation Equivariance - A feature learned by the CNN in the upper left-hand corner of a satellite image will be correctly identified regardless of where it appears in that satellite image (e.g., a roof feature). This is an important property of CNNs when dealing with very large, continuously updated geospatial raster images.

Sensitivity to Small Objects - CNNs' induction for local neighborhood connectivity makes them well-suited for performing dense prediction tasks for small objects, such as counting vehicles and extracting roads while keeping local edge detail intact; this way, CNNs can correctly identify some geospatial assets when the context is global in nature [12].

Structural Limitations of Foundation Models:

In spite of their ability to operate in ways that are useful to most users, CNNs have certain structural limitations in operation that prevent them from being useful as general-purpose foundation models.

Limited Global Context: CNNs accumulate data over an increasingly large area of visual space as they build receptive fields. Working with long-range dependencies such as modeling the continuation of a river segment from one corner of a map to another requires many layers (very deep) before creating a global way to reason about those locations. Compared to other mechanisms, like attention mechanisms, CNNs are indirect and computationally inefficient in constructing global reasoning about how to relate long-distance visual areas [2].

Task-Specific Representations: CNN representations are typically tightly coupled with the training objective/tasks that the CNN was trained on (e.g., image segmentation versus image classification). This tight coupling makes it difficult to learn a single, "universal," world model that can be transferred with zero-shot capability to other modalities or other resolutions.

Architectural Rigidity: CNNs typically expect to receive image input in fixed grid format. Therefore, CNNs do not have a natural means of working with irregular/non-grid-based data that

are frequently produced from multi-sensor fusion (an example would be LiDAR point clouds). They are being merged with optical rasters and do not provide a natural means of processing temporal data efficiently (variable-length time series) without complex recurrent connections [12].

TABLE 1. Architectural Trade-offs: CNNs vs. Vision Transformers

Property	CNNs (The Predecessor)	ViTs (The Foundation)
Inductive Bias	Strong (Local continuity & Equivariance)	Weak (Must be learned from data)
Global Context	Indirect (Limited by receptive field)	Direct (Global Self-Attention)
Small-Object Sensitivity	High (Preserves local signals)	Lower (May smooth out tiny details)
Scalability as FM	Limited (Saturates with depth)	High (Scales with data & compute)
Input Flexibility	Rigid (Fixed Grid)	Flexible (Token-based)

These limitations of CNNs motivated the research and development of architectures that are able to show explicit global reasoning and provide flexible token-based representations. This is the path that has led to the establishment of Vision Transformers as the current dominant architecture for modern remote sensing foundation models. A comparative summary of CNNs and Vision Transformers is provided in Table 1, highlighting their differences in inductive bias, scalability, and ability to model global context.

3.2 The Modern Standard: Vision Transformers (ViT)

Vision Transformers (ViTs) have become the prevalent backbone of modern foundation models in remote sensing due to their capability to model global spatial context via self-attention [7]. In contrast to CNNs, which combine data from a locally growing receptive field, ViTs work with

tokenized image patches that allow for direct scenewide interactions through all layers in the network. This property is particularly relevant in Earth observation applications where the semantic understanding of what is present often depends on long-range (between objects) relationships, such as how the connection between the river systems across the full extent of a given area or urban expansion across an entire city all have to be contextualized separately from the convolutional kernel's local context [70].

Core Architecture:

ViTs are architecturally aligned with the foundation model paradigm. The vision-transformer architecture uses a satellite image as the input to the model; it takes the input, breaks the image out into a series of fixed-size patches (usually $16 * 16$ pixels), and converts them to a linear token embedding using an embedding matrix. Since the transformer architecture is not spatially ordering, the positional encoding information is added to the token's embedding to provide some form of spatial ordering for tokens. Tokens are then transformed through MSA (Multi-Head Self-Attention) blocks, allowing each token to access all other tokens simultaneously in the sequence [7]. The vision-transformer architecture allows a model to learn flexible representations using data-driven methods rather than relying on CNN's hard-coded assumptions about the localities of the inputs by separating the spatial scale of an input from the depth of the architecture. Fig. 3 demonstrates how satellite images are decomposed into patch tokens and processed through self-attention layers, enabling direct global interaction between spatial regions. This allows the model to capture long-range dependencies across large-scale scenes, which is critical for remote sensing applications.

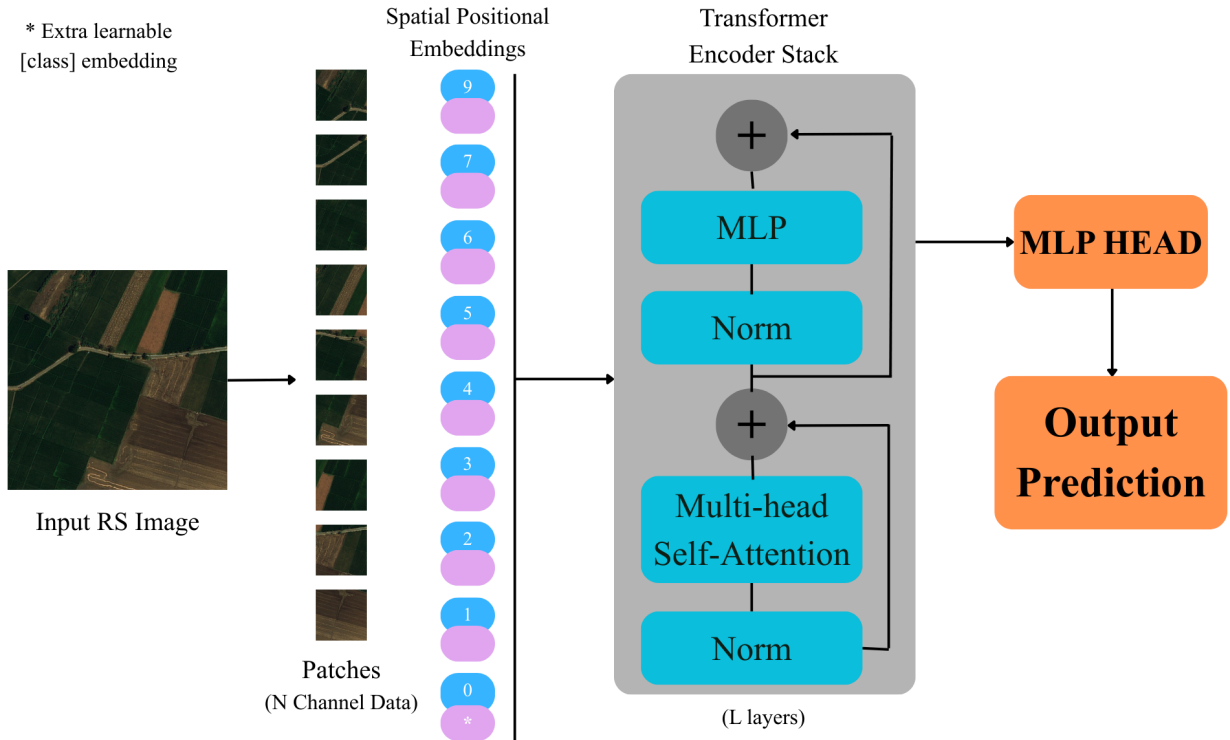


Fig. 3. Vision Transformer architecture. An input image is decomposed into fixed-size patches, embedded as tokens with positional encodings, and processed through stacked self-attention layers. Global attention enables direct modeling of long-range spatial relationships, a key advantage over convolutional architectures for large-scale satellite imagery. (Adapted from: [7])

Why ViTs work well as foundation model backbones:

ViTs are ideal as foundation model backbones, because they can be easily scaled. ViT representational capacity scales smoothly with both the available data and the compute resources, while also avoiding the saturation seen with CNNs [71].

Modality-agnostic tokenization: Because ViTs have a simple representation of an input to create a token from spatial grids and process using generic vectors, they are able to easily ingest multiple types of data. For example, the representation of a synthetic aperture radar (SAR) patch is structurally identical to that of an optical patch or a multispectral patch, making it possible to do the unified multi-sensor fusion that was demonstrated in Prithvi-EO-2.0 [5] and AnySat [55].

Pre-training Compatible: The token-based representation of both input and output fits naturally with the Masked Autoencoding (MAE) [4] approach, which is currently considered to be the gold standard in self-supervised learning. By randomly masking 75% of the input tokens, the computational cost is reduced while at the same time forcing the model to learn the global context required to reconstruct the masked portion of the input; it is less intuitive to create a global context in dense, convolutional grids [70].

4. KEY PRE-TRAINING STRATEGIES FOR ViT

In the context of foundation models, the "intelligence" of a vision transformer is largely determined by its pre-training objective. Pre-training plays a crucial role in the foundational models, where models are initially trained to learn generalized features and then fine-tuned on specific downstream tasks. The pre-training of ViT relies on self-supervised learning (SSL) [74] because labeled data is limited compared to abundant raw remote sensing data. SSL algorithms automatically generate supervisory signals from the data itself, allowing models to learn robust spatial, spectral, and temporal representations without human annotation. In the RS field, research is rapidly advancing, with major breakthroughs in both datasets and methods.

The current landscape of RS foundation models is dominated by three core methodologies: Masked Image Modeling (MIM) [13], Contrastive Learning (CL) [75], and Knowledge Distillation [15].

4.1 Masked Image Modeling (MIM)

MIM has emerged as the most dominant pre-training strategy for ViTs in remote sensing. The MIM process involves masking some information, e.g., pixels, patches, or even latent representations, and training a model, usually an autoencoder, to predict the missing information by using the context available in the visible part of the input [13].

1. Core mechanism: AutoEncoder

MIM models such as Masked Autoencoder (MAE) [4] typically follow a reconstruction-based framework. Fig. 4 illustrates the reconstruction-based MIM pipeline, where only visible patches are processed by the encoder, significantly reducing computational cost. This design forces the model to infer missing regions using global contextual understanding rather than relying on local pixel continuity.

The input image is first divided into a grid of non-overlapping, equally sized patches. A masking function is applied to the patched image, marking a significant portion of patches as "masked" while the remaining as "visible." The ViT [7] encoder processes only the visible patches, temporarily discarding the masked patches, to generate the latent representation. This makes the process computationally efficient. The learned representations are arranged into their original position, along with a learnable mask token for missing positions. The decoder processes this full sequence to reconstruct original pixel values of the missing area. The model minimizes the Mean Squared Error (MSE) between the reconstructed pixels and the original raw pixels, forcing the model to infer the missing information rather than simply memorizing the visible input [13].

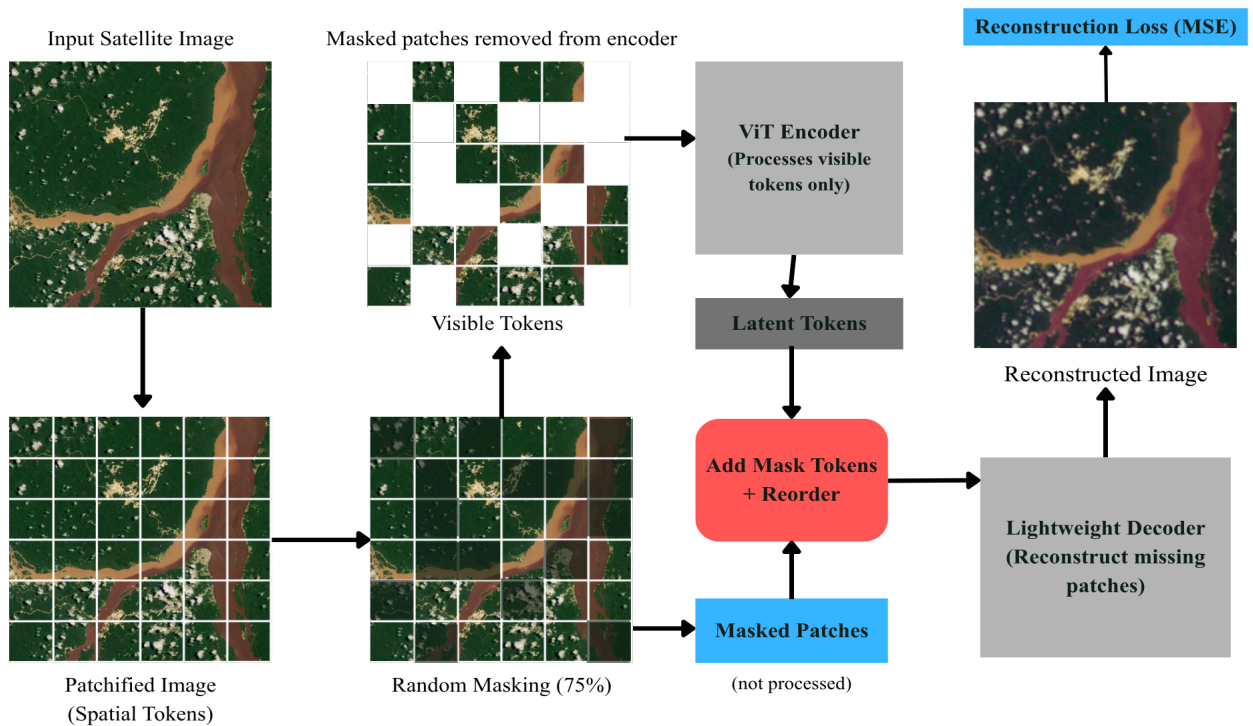


Fig. 4. Reconstruction-based Masked Image Modeling (MIM) framework. The input satellite image is divided into patches, and a subset is randomly masked. The encoder processes only visible patches, while a decoder reconstructs the original image using latent representations and mask tokens. The model is trained using a reconstruction loss to learn robust representations. (Adapted from: [13])

2. RS-Specific Variants

Standard MIM (like MAE [4]) treats images as static 2D snapshots. RS Foundation Models adapt this for geospatial complexities.

- Temporal Masking (SatMAE [17]): Satellite data is often a time series. SatMAE treats the input as a 3D volume. It masks tokens across the temporal dimension, forcing the model to predict how a landscape changes over time (e.g., distinguishing seasonal crop cycles from permanent land cover changes).
- Scale-Aware Masking (Scale-MAE [23]): Standard ViTs struggle with the varying zoom levels of satellite imagery. Scale-MAE incorporates Ground Sample Distance (GSD) into the positional embedding and inputs patches from different scales. It teaches the model that a "16x16 patch" represents different physical areas depending on the sensor resolution.
- Object-Oriented Masking (RingMo [24]): Random masking might miss small, critical objects like vehicles. RingMo employs a strategy to mask salient regions, forcing the model to "hallucinate" or reconstruct specific objects, which improves performance on downstream detection tasks.

4.2 Contrastive Learning (CL)

While MIM focuses on internal reconstruction, contrastive learning focuses on invariance and discrimination. Contrastive Learning (CL) is a self-supervised learning technique that trains models to distinguish between similar and dissimilar data points without requiring human-provided labels.

1. Core Mechanism: Feature Space Alignment

CL models such as SimCLR [75] and MoCo [76] typically follow a discrimination-based framework. The main aim of CL is to map data into low-dimensional representation (embedding) space where similar pairs are brought together and different pairs are pushed apart. Fig. 5 shows the contrastive learning framework, where augmented views of the same image are mapped close in feature space while dissimilar samples are pushed apart. This enables the model to learn invariant representations robust to rotation, illumination, and seasonal variations in remote sensing data.

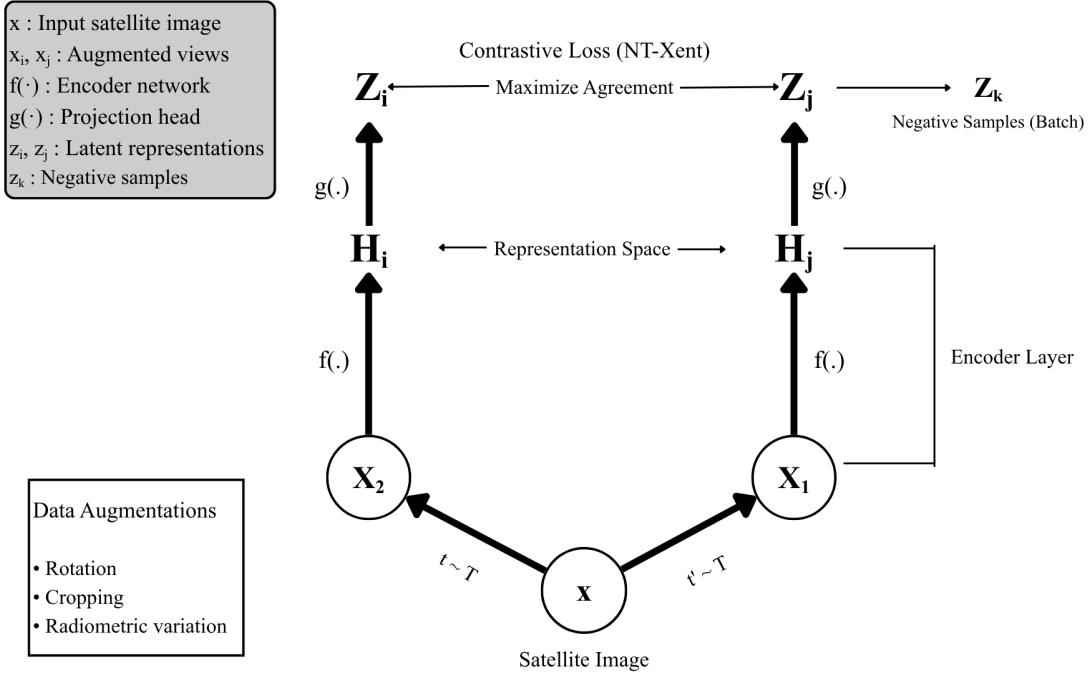


Fig. 5. SimCLR-based contrastive learning framework. Two augmented views of the same satellite image are generated using stochastic transformations and encoded through a shared encoder $f(\cdot)$, followed by a projection head $g(\cdot)$. The model maximizes agreement between positive pairs while minimizing similarity with negative samples within the batch using a contrastive loss function. (Adapted from: [75])

It operates through a two-step process, generating positive/negative pairs and optimizing a contrastive loss function. Since data is unlabeled, the system generates two different augmented versions (e.g., via random rotation, color jittering, or cropping) for a given image. The generated views/versions are similar or the “positive pairs.” Other, randomly selected images in a batch are used as negative samples. The model learns to differentiate the main image from these unrelated items. A contrastive loss maximizes the similarity between positive pairs while pushing negative pairs away [14]. For example, satellite imagery has no canonical orientation (no “up” or “down”). CL teaches the model that a rotated ship is still a ship (rotational invariance) or that a forest in shadow is the same as a forest in sunlight (radiometric invariance).

2. RS-Specific Variants

- Seasonal Contrast (SeCo) [77]: In natural images, “positive pairs” are made by artificial data augmentation. In RS, we can use the time axis. SeCo utilizes images of the same geographic location taken at different times as positive pairs. This forces the model to learn features that are invariant to seasonal changes (e.g., snow vs. green vegetation) while remaining sensitive to land cover identity.
- Tile2Vec [78]: An early precursor that uses geographic proximity; patches physically close to each other are treated as positive pairs, assuming spatial autocorrelation.

4.3 Knowledge Distillation & Hybrid Approaches

Recent state-of-the-art models often combine aspects of reconstruction and discrimination or use “teacher-student” architectures to stabilize training. Knowledge Distillation (KD) [15] is a technique for transferring knowledge from a large teacher model to a smaller, efficient student model to help improve performance on limited-resource devices. Hybrid approaches enhance this by combining KD with techniques like feature mapping or attention mechanisms, often achieving superior speed and accuracy. Key examples include multi-level distillation, teacher-driven data generation, and hybrid attention transfer. Fig. 6 demonstrates the self-distillation mechanism, where a student network learns from a teacher updated via an exponential moving average. This setup stabilizes training and enables the model to learn rich semantic representations without explicit labels.

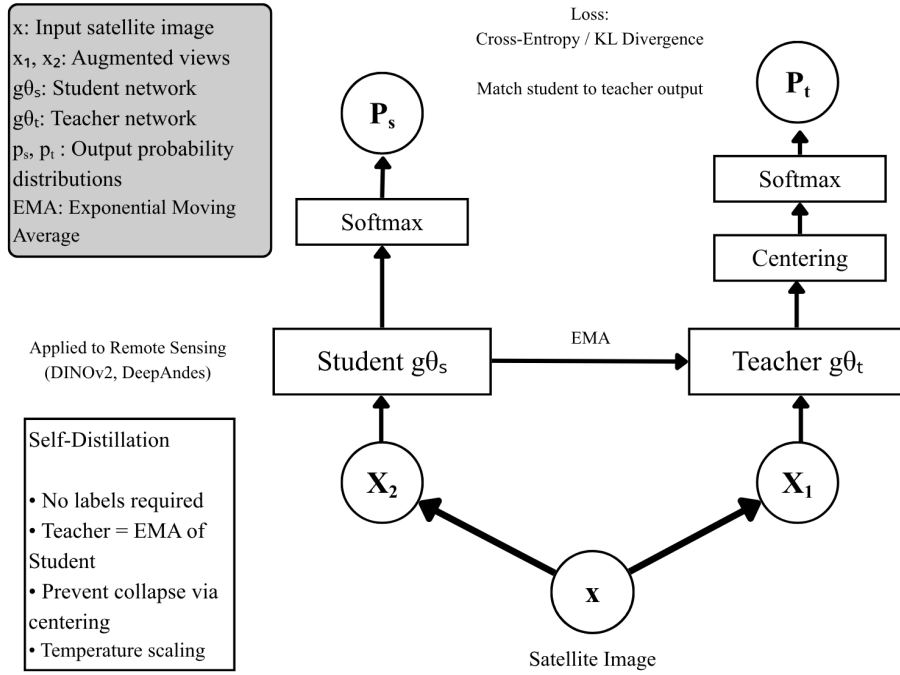


Fig. 6. DINO self-distillation framework. Two augmented views of the same satellite image are processed by a student and a teacher network. The teacher network is updated as an exponential moving average of the students. The student is trained to match the teacher's output distribution using a cross-entropy loss, while centering and temperature scaling are applied to prevent collapse. [Adapted from [65]]

1. Core Mechanism: Self-Distillation (DINO)

Methods like DINO [65] (Distillation with NO labels) comprise two ViT architectures: a student and a teacher. A large pre-trained model (teacher) is used to train a smaller compact model (student). The student learns the patterns by matching the output of the teacher's patterns with the help of exponential moving averages. The teacher provides soft labels that are probabilities rather than just providing hard labels that contain dark knowledge, rich information of the data structure. Temperature scaling is used to avoid the mode collapse where the model provides the same vector for everything. This helps the model to learn high-quality semantic features without the need for manual labeling, making it highly effective for image processing.

2. RS Implementation

- DeepAndes [79]: The DeepAndes foundation model explicitly utilizes the DINOv2 [16] framework. The authors note that DINOv2 aligns well with the "dense prediction" nature of RS tasks (like semantic segmentation). Unlike CL, which often works on global image vectors, DINOv2 encourages local patch-level understanding, which is vital for identifying small archaeological features in complex terrains.
- SkySense [80]: While primarily a multi-modal model, it employs a massive unified contrastive framework (Geocontextual Prototype Learning) that acts similarly to a hybrid approach, aligning optical and SAR data in a shared space.

Each pre-training strategy exhibits distinct strengths and limitations depending on the target task and data characteristics. Reconstruction-based methods prioritize spatial fidelity, and contrastive approaches emphasize invariance, while distillation-based methods enhance efficiency and robustness. These differences are summarized in Table 2, which compares their core mechanisms, strengths, limitations, and suitability for different remote sensing tasks.

TABLE 2. Comparison of the Pre-training strategies

Strategy	Core Mechanism	Key Strengths (+)	Key Limitations (-)	Optimal Tasks	Examples
MIM	Reconstruction : Restores masked image patches.	Captures fine texture & spatial structure.	Weak global semantic abstraction.	Dense Prediction (Segmentation)	SatMAE, Prithvi
CL	Discrimination : Aligns augmented views.	Robust to seasonal & lighting changes.	Sensitive to augmentation choices.	Scene Classification (LULC)	SeCo, Tile2Vec
KD	Distillation:	High noise	Performance	Model	DINO,

	Student mimics Teacher.	tolerance & compactness.	limited by Teacher quality.	Compression & Transfer	SkySense
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The scale, diversity, and modalities of the datasets used to pre-train foundation models have a significant impact on their performance in remote sensing. This enables them to learn about the spatial, spectral, and temporal variability of a geographical region or type of sensor when there are large volumes of data to train on. The various key datasets used for pre-training foundation models are summarized in Table 3; all datasets include an indication of their modality/level of supervision and most likely type of problem they were used to solve.

TABLE 3. Datasets for Pre-training Remote Sensing Foundation Models

Dataset	Primary Modality	Key Property	Supervision	Primary FM Type
fMoW	Optical	Global-scale functional semantics	Strong labels	Vision
SEN12MS	Optical + SAR	Aligned multi-sensor imagery	Unlabeled	Vision / Multimodal
BigEarthNet-MM	Optical + SAR	Large-scale multi-label semantics	Weak labels	Vision / Multimodal
MillionAID	Optical (Aerial)	Diverse scene categories	Strong labels	Vision

SSL4EO-S12	Sentinel-1/2	Multi-modal, multi-temporal scale	Unlabeled	Vision / Spatiotemporal
SSL4EO-L	Landsat	Long-term temporal coverage	Unlabeled	Vision / Spatiotemporal
SatlasPretrain	Optical + OSM	Massive weak supervision	Weak labels	Vision / Multimodal
SeeFar	Optical (Multi-GSD)	Resolution-agnostic pretraining	Unlabeled	Vision
RS-4M	Optical	Extremely large-scale MAE pretraining	Unlabeled	Vision
Major TOM	Multi-sensor	Expandable global EO corpus	Mixed	Vision
Copernicus-Pretrain	Sentinel family	Unified European EO pretraining	Unlabeled	Vision / Spatiotemporal
EarthNet2021	Optical (Time Series)	Earth digital twin forecasting	Strong labels	Spatiotemporal
GeoPile	Optical + Text	Curated VL pretraining corpus	Weak labels	Vision–Language
RSVG	Optical + Text	Visual grounding	Strong labels	Vision–Language
RS5M	Optical + Text	Large-scale image–text pairs	Weak labels	Vision–Language

SATIN (Metadataset)	Optical + Text	Multi-task VL supervision	Strong labels	Vision–Language
SkyScript	Optical + Text	Semantically rich captions	Weak labels	Vision–Language
ChatEarthNet	Optical + Text	Global-scale VL alignment	Weak labels	Vision–Language

5. Foundational Model in RS

Foundational models are pre-trained models on broad types of data, such as geospatial images, which often use self-supervised learning, which acts as a base of foundational models. It is able to adapt multiple downstream tasks [3]. Earlier, CNN workflow was task-specific. If a researcher wanted to detect ships, they specifically needed to train a model for ships; if it was detecting different types of landforms, they had to make a separate model for it. The approach was inefficient, rigid, and used a lot of computational resources.

Whereas foundational models are task agnostic. They are pre-trained on abundant data, like satellite imagery, so these models understand the universal Earth surface. It learns the physics of remote sensing, understanding different rivers, menders, urban areas, vegetation, and rigid geometrics without telling them explicitly about these objects [3]. As these models have pre-existing knowledge, they can be fine-tuned on specific and complex problems like road detection, cloud segmentation, and change detection without giving them additional information and providing less labeled data than CNNs.

The "Pre-train, Fine-tune" Lifecycle:

The lifecycle of the foundational model discards the traditional “train from scratch” method, splitting the learning process into two distinct phases: pre-training and fine-tuning. The first phase, pre-training, is a computationally intensive step where the model is exposed to massive unlabelled data such as Sentinel-2. The main objective is to learn a generalized representation of

the world. It results in a set of model weights that captures the fundamental statistical properties of the geospatial data.

Once the initial phase ends, fine-tuning begins. The second phase is extremely efficient, which requires only a fraction of computational power, which can be achieved on a single GPU also. In this phase the foundational model is specifically tuned to a specific dataset using a small labeled exam. As the model already understands the structure of the landscape, it does not require re-learning how to see it, which results in higher accuracy, producing a specific special model at less cost and time.

Key Characteristics of Remote Sensing Foundation Models:

The section distinguishes foundational models from earlier pre-trained models by defining three characteristics: emergence, homogenization, and generalizability.

1. Emergence and Scaling Laws:

Earlier models hit a performance plateau, while foundation models followed scaling laws, meaning their potential improved predictably and indefinitely as the model size and dataset size increased. This provides empirical evidence that increasing model size (from 100M to 600M+ parameters) on Harmonized Landsat-Sentinel (HLS) data directly correlates with lower error rates in downstream tasks. It is the proof that scaling laws apply to Earth observation. More importantly, they exhibit emergent properties, skills that the model was never explicitly taught. [3] says that “the behavior of a foundational model is an emergent property, meaning it is implicitly induced rather than explicitly constructed.” For example, a foundation model is only trained to predict the weather for the next day. To do that really well, it might 'teach itself' how to recognize different cloud types and wind patterns, even though no one ever specifically asked it to be a cloud or wind expert. It develops that skill naturally because it's a useful stepping stone to its main job of forecasting.

2. Homogenization:

Foundation models offer a revolutionary and unified framework for processing heterogeneous data types. Earlier, optical, SAR (Radar), and LiDAR [10] data required different, incompatible algorithms. FMs solve this by treating all data inputs as generic "tokens" in a shared embedding space. This allows for cross-modal transfer, where the model can utilize its rich understanding of abundant optical imagery to better interpret sparse, difficult-to-label SAR data, effectively translating knowledge across different physical sensors [3].

3. Generalizability (Zero-Shot and Few-Shot):

The ultimate aim for foundation models is to make them tough enough to handle different situations, say, the scenery or time of year changes (like when a model trained on summer photos completely fails in the winter). Because foundation models have "seen" the Earth under every conceivable lighting condition, angle, and season during pre-training, they learn invariant features, core truths about objects that don't change with the weather [17]. This deep robustness enables Few-Shot Learning, allowing the model to master a new class with just a handful of examples, or even Zero-Shot Learning, where it can perform tasks it has never seen before, guided only by simple prompts.

5.1 TYPES OF FOUNDATIONAL MODELS

1. VISION FOUNDATIONAL MODELS

Vision Foundational Models (VFMs) lay the “backbone” of the modern remote sensing applications. Defined as deep learning models pretrained on massive, diverse remote sensing datasets, designed to extract universal, generalizable visual features [3]. VFMs learn invariant features that can be effectively used to perform multiple downstream tasks with minimal fine-tuning, unlike earlier task-specific models such as U-Net [18], which were trained exclusively for road detection on small datasets.

While the underlying architectures utilize Vision Transformers (ViT) [7] and Swin Transformers [19] (discussed in Section 3), the training methodologies rely on Self-Supervised Learning (SSL) strategies like Masked Image Modeling (MIM) [13] and

Contrastive Learning (CL) [14] (discussed in Section 4), the true innovation of VFMs lies in their application at scale. By utilizing massive datasets such as BigEarthNet [20], fMoW (Functional Map of the World) [21], and Million-AID [22], these models address the historic scarcity of labeled data. This scaling allows them to achieve state-of-the-art performance even in few-shot scenarios, where only a handful of labeled examples are available.

State-of-the-Art Models:

Right now, the research really points to some groundbreaking Vision Foundation Models (VFMs) that show just how well standard computer vision approaches can be adapted to handle the unique difficulties of satellite and aerial imagery.

- **MIM-Based Models (SatMAE & Scale-MAE):** Models that use Masked Image Modeling are incredibly good at tasks that require detailed, pixel-level predictions. SatMAE [17] adapts the Transformer architecture to treat time as a third dimension, capturing phenological trends by masking patches across temporal sequences. Meanwhile, Scale-MAE [23] addresses the multi-resolution challenge inherent to satellite imagery (e.g., comparing Sentinel vs. Dove data) by introducing a scale-aware positional encoding, allowing the model to process data ranging from 0.3 m to 30 m resolution without degradation.
- **Generative Models (RingMo):** RingMo [24] represents the first generative self-supervised foundation model specifically for remote sensing. Trained on 2 million images, it employs a novel strategy to mask salient objects (rather than random background patches). This forces the model to "hallucinate" or reconstruct complex missing features, resulting in superior performance for detecting small objects like ships or vehicles.

Despite their success, challenges remain regarding multimodal fusion and computational scalability. However, the field is rapidly evolving toward the integration of Vision-Language Models (VLMs). Emerging frameworks like RemoteCLIP [25] allow for natural language queries (e.g., *"show me flooded areas"*), enabling zero-shot inference

and significantly enhancing the accessibility of these complex models for non-technical users. To better understand the design space of vision foundation models, Table 5 presents a structured comparison of representative approaches, emphasizing how different architectures and pre-training strategies address challenges such as multi-scale variability, temporal dynamics, and multi-modal fusion in remote sensing.

TABLE 5. Prominent Vision Foundation Models In Remote Sensing

Model	Architecture	Pre-training Strategy	Input Modality	Key Capability
I. Optical & Spatiotemporal Models				
SeCo	ResNet / ViT	Seasonal Contrastive	Optical (Temporal)	Seasonal invariance (e.g., snow vs. green).
SatMAE	ViT-Large	Masked Image Modeling	Optical (MS + Time)	Strong spatial–temporal representations.
SatMAE++	ViT	Multi-Scale MAE	Optical (MS)	Optimized multispectral pre-training.
RingMo	ViT	Rotational Masked Modeling	Optical (High-Res)	Object-centric learning for oriented targets.
Scale-MAE	ViT	Scale-Aware MAE	Optical (Multi-GSD)	Resolution-robust encoding (0.1–30m).

CACo	ViT	Change-Aware Contrastive	Optical (Temporal)	Change-sensitive features.
Prithvi-EO-1.0	ViT	Spatiotemporal MAE	Optical (HLS)	General-purpose EO encoding.
Prithvi-EO-2.0	ViT-Huge	ST-MAE + Metadata	Optical (MS + Time)	SOTA transfer with explicit metadata.
RS-4M	ViT	Large-Scale MAE	Optical	Billion-scale pre-training.
SeeFar	ViT	Multi-Resolution SSL	Optical	Sensor-agnostic vision encoding.
SatMamba	Mamba (SSM)	Self-Supervised Sequence	Optical	Linear-complexity large-scene modeling.
Copernicus-FM	ViT	SSL Pre-training	Sentinel Family	Unified European EO modeling.
II. Multi-Modal (Optical + SAR) Models				
CROMA	ViT	Radar–Optical MAE	SAR + Optical	Joint representations via cross-attention.
USat	ViT	Unified SSL	Multi-sensor	Cross-sensor generalization.
SkySense	Swin-L	Multi-task SSL	Multi-modal EO	Universal EO interpretation.

MMEarth	ViT	Multi-modal Pretext Tasks	Optical + SAR	Cross-modal representation learning.
OmniSat	ViT	Modality-Fu sion SSL	Optical + SAR	Modality-agnostic encoding.
III. Hyperspectral Models				
DOFA	ViT	Frequency- Adaptive MAE	Optical (MS/Hyper)	Spectral generalization via dynamic weights.
SpectralGPT	Transformer	Spectral Token Modeling	Hyperspectral	Captures continuous spectral signatures.
S2MAE	ViT	Spatial–Spe ctral MAE	Hyperspectral	Spectral–spatial coupling.
SpectralEarth	Transformer	Large-scale HSI Pretraining	Hyperspectral	Scalable hyperspectral FM.
HyperSIGMA	Transformer	Hyperspectr al SSL	Hyperspectral	Large-scale HSI foundation modeling.
HyperFree	Transformer	Channel-Ad aptive SSL	Hyperspectral	Tuning-free HSI transfer.

2. VISION LANGUAGE MODEL

Vision-language models (VLMs) transition to a high level of “semantic reasoning” in remote sensing. While VFMs extract features for predefined classes, they are limited by the fixed vocabulary of training data. VLMs overcome this by aligning visual inputs with natural language [26], enabling open-vocabulary tasks. This capability allows users to interact with geospatial data using unstructured text, such as querying *"Find all coastal areas with significant mangrove degradation,"* without requiring a specific classifier trained exclusively on "mangrove degradation" [27].

The core innovation of VLMs lies in their ability to map two distinct modalities, satellite imagery and human language, into a shared, high-dimensional embedding space [28]. This enables the model to understand that the visual pattern of a "white cylindrical tank" is semantically equivalent to the textual description "oil storage facility," allowing for seamless translation between pixel data and human concepts. VLMs utilize a Dual-Encoder or Fused-Encoder architecture. This typically involves a Vision Transformer (e.g., ViT-L/14) to encode the image into visual tokens and a Text Transformer (e.g., BERT [29]) to encode the textual prompt.

State-of-the-Art Models:

- RemoteCLIP [25]: The first true foundation model for RS vision-language tasks. By scaling pre-training to 12 heterogeneous datasets (including data derived from object detection boxes converted to captions), it achieves state-of-the-art performance in zero-shot classification, outperforming standard OpenAI CLIP by significant margins on geospatial benchmarks.
- RSGPT [30]: A generative pre-trained transformer specifically tailored for remote sensing. Unlike retrieval models that only *find* images, RSGPT can *generate* detailed descriptive captions for complex scenes (e.g., *"A dense residential area with varying roof colors located near a river bank"*), facilitating automated report generation.
- GeoChat [31]: The first "grounded" large vision-language model for remote sensing. It addresses a critical limitation of generic VLMs: the inability to localize

objects. GeoChat can accept region-specific queries (e.g., *"What is inside this bounding box?"*) and return spatially grounded responses, making it highly effective for precise surveillance and monitoring tasks. A structured comparison of prominent vision-language models in remote sensing is presented in Table 6, highlighting differences in training paradigms, backbone architectures, and their capabilities in tasks such as zero-shot classification, visual grounding, and multimodal reasoning.

TABLE 6. PROMINENT VISION-LANGUAGE MODELS IN REMOTE SENSING

Model	Vision / Lang. Backbone	Training Paradigm	Key Capability
I. Contrastive Vision-Language Models (CLIP-Style)			
RemoteCLIP	ViT / CLIP Text	CLIP-style Contrastive Alignment	Robust zero-shot classification & retrieval.
GeoRSCLIP	ViT / CLIP Text	Large-Scale Pre-training (RS5M)	Scalable vision–language alignment.
RSCLIP	ViT / CLIP Text	Weakly Supervised Alignment	Human-annotation-free VL learning.
GRAFT	ViT / Transformer	Annotation-Free Ground–Image Alignment	Label-free pre-training via overhead imagery.
DOFA-CLIP	ViT / CLIP Text	Frequency-Aware Alignment	Cross-modal generalization across sensors.
II. Generative & Instruction-Tuned Models (LLM-Based)			

GeoChat	ViT / LLaMA	Grounded Instruction Tuning	Interactive multi-turn RS reasoning.
RSGPT	ViT / LLM	Instruction Tuning	Generalization to diverse RS tasks.
RS-LLaVA	ViT / LLaVA-LLM	Joint Captioning & VQA	Unified visual reasoning.
SkySense	Swin / LLM	Fine-Grained Instruction Tuning	High-level semantic interpretation.
SkySense-O	ViT / LLM	Open-World VL Modeling	Open-vocabulary interpretation.
Falcon	ViT / LLM	Unified VL Foundation	General-purpose RS reasoning.
RSUniVLM	ViT / MoE-LLM	Granularity-Aware Fusion	Multi-scale understanding via MoE.
RingMo-Sense	ViT / LLM	Multi-Modal Tuning	Unified vision–language–grounding.
VHM	ViT / LLM	Robust Alignment	Trustworthy & hallucination-free reasoning.
III. Grounding & Pixel-Level Models			
RSVG	CNN/ViT + Bert	Visual Grounding Supervision	Region-level object localization from text.

EarthMarker	ViT / Transformer	Visual Prompt Learning	Precise point & region comprehension.
GeoGround	ViT / LLM	Unified Grounding Framework	Spatial grounding of complex queries.
LISat	ViT / LLM	Language-Instructed Segmentation	Text-driven segmentation masks.

3. GENERATIVE FOUNDATION MODELS

Generative Foundations Models in remote sensing transition from analyzing to creating new data [32]. While VFM are discriminative, Generative FMs generate realistic satellite imagery by learning the probability distribution of Earth data. These models are designed to solve "Inverse Problems", restoring corrupted data (e.g., cloud removal), enhancing quality (super-resolution), or simulating realistic training data from textual descriptions (text-to-image).

The design of Generative Foundation Models in remote sensing has decisively shifted from Generative Adversarial Networks (GANs) [33] to Denoising Diffusion Probabilistic Models (DDPMs) [34] due to the latter's robustness to instability and mode collapse. In the Diffusion Framework, the model is trained to reverse a noise process step by step, purifying a Gaussian noise signal into physically valid satellite images conditioned on text inputs. Most importantly, to guarantee geospatial correctness rather than artistic validity, "zero-convolution" layers are used in ControlNet-RS [35], which forces the generation process to strictly follow certain spatial configurations (e.g., road maps) and metadata, guaranteeing that the generated data is geospatially accurate.

State-of-the-Art Models:

- DiffusionSat [36]: Currently the largest generative foundation model for satellite imagery. It is trained on massive datasets to perform multiple downstream tasks, including super-resolution and temporal prediction. Uniquely, it uses metadata

conditioning, allowing users to generate images based on specific years or Ground Sampling Distances (GSD).

- CRS-Diff (Controllable Remote Sensing Diffusion) [37]: A framework specifically designed to integrate ControlNet with satellite data. It allows for "decoupled" generation, where a user first generates a semantic map (the layout) and then uses the diffusion model to paint realistic textures onto that map, ensuring perfect alignment between the image and its label. Table 7 summarizes prominent generative foundation models in remote sensing, highlighting how different generative paradigms such as diffusion and conditional modeling enable tasks including data synthesis, super-resolution, and controllable geospatial simulation.

TABLE 7. Prominent Generative Foundation Models In Remote Sensing

Model	Generative Paradigm	Conditioning/Input	Key Capability	Typical Applications
DiffusionSat	Latent Diffusion	Optical/Multispectral	Large-scale satellite image generation.	Data augmentation, representation learning.
MetaEarth	Diffusion Model	Optical (Global-Scale)	Unbounded global-scale EO synthesis.	Simulation, large-area synthesis.
CRS-Diff	Diffusion Model	Controllable Attributes	Attribute-conditioned generation.	Scenario simulation, controllable synthesis.
Text2Earth	Diffusion Model	Text + Vision	Text-driven EO image synthesis.	Scenario generation, zero-shot simulation.

Seg2Sat	Conditional Diffusion	Segmentation Masks	Layout-to-image synthesis.	Annotation bootstrapping, urban planning.
HSIGene	Diffusion/Transformer	Hyperspectral	Hyperspectral data generation.	HSI augmentation, spectral analysis.
GenScotland	Conditional GAN/Diffusion	Maps / Geographic Priors	Region-specific satellite synthesis.	Climate adaptation planning.

4. VISION-LOCATION FOUNDATION MODELS

Vision-Location Foundation Models address a fundamental limitation of standard computer vision: Spatial Agnosticism. A standard deep learning model sees a patch of "green pixels" and cannot distinguish whether it is a tropical rainforest in Brazil or a boreal forest in Canada. Vision-Location FMs explicitly integrate geospatial coordinates (latitude/longitude) and time as core inputs alongside the image pixels [81]. By learning the statistical relationship between "where" an image is taken and "what" it contains, these models effectively encode the First Law of Geography ("everything is related to everything else, but near things are more related than distant things"), allowing for spatially aware predictions that standard models cannot achieve.

Vision-Location Foundation Models relies on the fusion of visual features with a learned location embedding to construct a unified, spatially aware representation. To overcome the spectral bias inherent in processing raw latitude/longitude coordinates, these architectures utilize high-frequency encoding schemes such as sinusoidal encoders, spherical harmonics, or CoordMLPs [38] to map geospatial positions into a high-dimensional vector space capable of resolving fine-grained spatial boundaries. This representation is optimized via contrastive geo-alignment, a training objective that minimizes the distance between the visual embedding of a scene and its corresponding coordinate embedding, effectively forcing the model to learn the statistical correlations between visual semantics (e.g., specific soil signatures) and their geographic distribution.

State-of-the-Art Models:

- SatCLIP [39]: A global, general-purpose location encoder pre-trained on Sentinel-2 imagery. It matches satellite images to their coordinates using a contrastive loss. The resulting embeddings capture not just location but also the *characteristics* of that location (e.g., climate, biome, urbanization level), effectively acting as a "smart GPS" that tells a downstream model what to expect at that coordinate.
- GeoCLIP [40]: A clip-inspired framework designed for worldwide geo-localization. It learns a continuous representation of the Earth, allowing it to retrieve the GPS coordinates of a query image by matching it against a gallery of learned location embeddings. It effectively creates a "mental map" of the entire planet's visual appearance. Table 8 presents a comparison of prominent vision-location foundation models, highlighting how spatial encoding strategies and geo-alignment mechanisms enable geographically aware representations and improve geo-localization and spatial reasoning tasks.

TABLE 8. Prominent Vision-Location Models in Remote Sensing

Model	Vision Backbone	Location Signal	Learning Paradigm	Key Capability
I. Spatial Contrastive & Neighborhood-Based Models				
Tile2Vec	CNN (ResNet)	Spatial Proximity	Spatial Contrastive Learning	Neighborhood-aware embeddings via Tobler's Law.
GeoKR	ResNet / ViT	Geo-Context + Coords	Geography-Aware SSL	Spatially grounded representations for LULC.

GASSL	ResNet	Geo-Coordinates	Geo-Aware Contrastive SSL	Location-consistent feature learning.
CSP	ResNet / ViT	Spatial Neighborhoods	Contrastive Spatial Pre-training	Geo-aware visual representations.
GeCo	ViT	Geo-Supervision	Geography-Gu ided SSL	Reduction of label bias in geo-representations
II. Coordinate Encoding & Geo-CLIP Models				
GeoCLIP	ViT	GPS Coordinates	Image–Locatio n Contrastive	Worldwide zero-shot geo-localization.
SatCLIP	ViT	Global Coordinates	CLIP-style Geo-Embeddin g	General-purpose global location embeddings.
EarthPT	Transformer	Spatiotemporal Coords	Large-Scale Pre-training	Earth-scale spatial representations from time-series.
RANGE	Transformer	Continuous Coords	Retrieval-Aug mented Neural Fields	Multi-resolution geo-embeddings.
III. Unified & Implicit Representation Models				
AnySat	ViT	Multi-Scale Alignment	Unified EO Pre-training	Location & sensor generalization (Geo-aware mode).

5. VISION-AUDIO FOUNDATIONAL MODELS

Vision-Audio Foundation Models represent the newest frontier in multimodal remote sensing, fusing overhead satellite imagery with ground-level acoustic data. Despite the physical inability of satellite sensors to detect sound, these models employ deep learning to establish a correlation between the visual characteristics of land cover and urbanization and their associated soundscapes. This enables the inference of ground-level acoustic activity such as traffic congestion, industrial humming, or bird biodiversity purely from orbital imagery [41]. By bridging the gap between the "view from above" and the "sound from below," these models offer a novel proxy for monitoring noise pollution and ecological health at a global scale.

Vision-Audio Foundation Models use cross-view cross-modal learning [42] to align satellite imagery with ground-level acoustic data in a shared latent space. They are trained on large-scale datasets of paired satellite images and geotagged audio (from sources like Freesound or BirdNET) to link visual land cover with environmental soundscapes. Architecturally, these models often employ Audio Spectrogram Transformers (AST) [43], which process audio as spectrograms, converting them into "Audio Tokens" (e.g., traffic sound) that are contrastively aligned with "Visual Tokens" (e.g., highway). This joint representation learning allows the model, during inference, to synthesize or retrieve likely acoustic profiles solely from visual satellite inputs in zero-shot scenarios, even with the audio branch discarded.

State-of-the-Art Models:

- Sat2Sound [44]: A pioneering framework that predicts fine-grained soundscapes from satellite imagery. It utilizes a zero-shot approach to generate realistic audio for any location on Earth, effectively creating a "global acoustic map." It creates a shared codebook of soundscape concepts, allowing it to retrieve or synthesize sounds (e.g., *wind in trees* vs. *wind in skyscrapers*) based solely on visual textures.

- ALAC (Adaptive Learning for Aligning Correlation) [42]: A retrieval-focused model that aligns remote sensing images with audio clips. It addresses the information asymmetry between the two modalities (images are content-rich; audio is abstract) by using region-enhanced attention, ensuring that specific visual objects (like a stadium) trigger the correct acoustic associations (like crowd noise).

Unlike VFMs, this category is still in the "Specialized Framework" stage and lacks a single dominant "Foundation Model" like GPT, but it is a critical part of the multi-modal future. Table 9 summarizes key research in vision-audio remote sensing, illustrating how cross-modal alignment between satellite imagery and acoustic signals enables novel applications such as soundscape prediction and environmental monitoring.

TABLE 9. Key Research in Vision-Audio Remote Sensing

Model / Dataset	Vision Modality	Audio / Signal Modality	Learning Paradigm	Key Capability
SoundingEarth	Aerial Imagery	Environmental Audio	Cross-Modal Contrastive	Unsupervised geolocation of ground audio using aerial imagery.
GeoBind	Satellite Imagery	Audio + Text	CLIP-style Binding	Unifying audio, ground imagery, and text via satellite anchors.
TFAVCNet	Aerial Imagery	Environmental Audio	Two-Stage Fusion	Audiovisual scene classification (e.g., industrial vs. residential).

DeepSense	Camera / Visual	Wireless / RF Signals	Multimodal Deep Learning	Co-sensing for telecom and urban environments (6G sensing).
QuakeGPT	(Seismic Focus)	Seismic Waveforms	Time-Series Forecasting	Earthquake nowcasting and sequence prediction.
Sat2Sound	Satellite (Multi-scale)	Soundscapes	Zero-Shot Mapping (VLM-based)	Synthesizes realistic audio for any location ("Global Acoustic Map").
ALAC	RS Imagery	Audio Clips	Adaptive Correlation Alignment	Region-aware alignment for retrieving audio from images (and vice versa).
ADVANCE	Aerial Imagery	Environmental Audio	Dataset / Benchmark	The standard benchmark for audiovisual remote sensing scene recognition.

5.2 KEY FOUNDATIONAL MODELS

The rapid evolution of remote sensing AI is best characterized by the development of specialized Foundation Models (FMs). The following section details ten models proposed between 2022 and 2026, selected for their architectural distinctiveness and contributions to solving specific

geospatial challenges. Table 10 provides a comparative summary of key remote sensing foundation models, highlighting differences in architectural design, pre-training objectives, data scale, and their unique capabilities across diverse geospatial tasks.

1. SatMAE (Temporal & Multi-Spectral MAE) [17]

SatMAE has a ViT-Large [7] base with 304 million parameters, and it has been adjusted to process non-RGB spectral data. SatMAE differs from a traditional vision network in how it does its Groupwise Spectral Embeddings. SatMAE groups spectral bands by their wavelength (i.e., RGB, NIR, and SWIR) and projects them independently along their own depth of the channel.

The model will be trained using masked autoencoding with 75% masking of the input data. The novel aspect of MAE [4] in this case is that tokens are also masked across the temporal dimension, which means that the model will be required to predict (i.e., reconstruct) missing time steps using temporal relationships (phenological trends) rather than solely spatial information.

The impact of the fMoW-Sentinel dataset [21] (pre-trained using ImageNet initialization) compared to the UCMerced dataset [45] was a 2.6% mIoU improvement with the use of datasets pre-trained on the v2 dataset versus ImageNet-initialized weights, providing evidence that spectral pre-training is necessary for domain-specific spectral representation.

The SatMAE framework is illustrated in Fig. 7, where the input satellite image is divided into patches and a large portion is randomly masked during training. As shown in Fig. 7, the encoder processes only visible tokens while a lightweight decoder reconstructs the masked regions to learn meaningful representations.

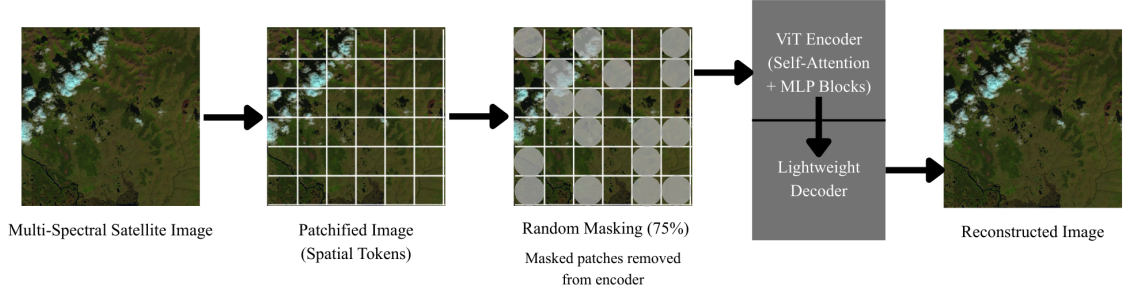


Fig. 7. SatMAE works as a powerful SSL vision learner, with its carefully designed masking strategies across multispectral and temporal images. (Adapted from: [17])

2. Scale-MAE (Scale-Aware Representation Learning) [23]

The multi-layer architecture of our ViT-base model [7] (with 86M weights) was specifically designed for the "multi-scale" problem where sizes of objects (for example, automobiles) appear drastically different as a result of sensor resolution, or GSD (Ground Sample Distance).

We pre-trained our model using a scale-aware MAE [4] and conditioned the input of the encoder on the GSD of the image through a GSD decoding process. For example, we took the feature space of each image and mapped them into a metric space (in meters as opposed to pixel units), thereby creating a basis to measure the size of each object.

Combining fMoW [21] and Million-AID [22] datasets (resolutions from 0.1 m through to 30 m), we have achieved the highest level of transfer performance across multiple scales; our model outperformed other standard ViTs by a large margin when measured against images of unknown resolution(s).

The Scale-MAE framework is illustrated in Fig. 8, where multi-resolution satellite images are encoded along with Ground Sample Distance (GSD) information to capture scale variations. As shown in Fig. 8, the model integrates scale-aware positional encoding into masked autoencoder training to learn scale-invariant representations.

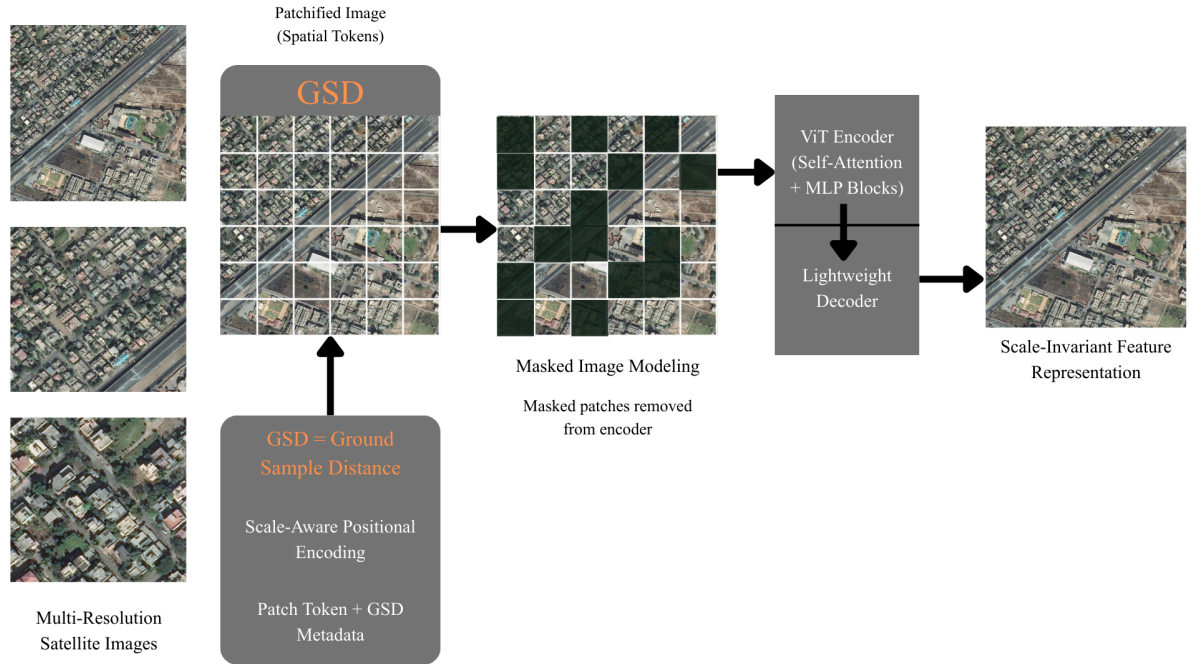


Fig. 8. Overview of the Scale-MAE framework for scale-invariant representation learning in remote sensing. Multi-resolution satellite images are patchified into spatial tokens, and Ground Sample Distance (GSD) metadata is incorporated through scale-aware positional encoding. The model is trained using masked image modeling, where visible tokens are processed by a Vision Transformer encoder and a lightweight decoder reconstructs masked patches to learn scale-robust feature representations. (Adapted from: [23])

3. Presto (Lightweight Timeseries Transformer) [46]

The Presto was tailored to be both small and efficient (the pre-trained model has ~20 million parameters) in terms of its architecture and scaling. In contrast to Vision Transforms (ViT) [7], which view the input as spatial patches, Presto treats the temporal history for each pixel as a sequence of tokens. Thus, creating a new class of Transformer called a Pixel Level Transformer.

The pre-training strategy utilized by Presto combines two different mechanisms, Masked Sensing Modeling, which will randomly mask out some subset of all time steps and

spectral bands, with geo-encoding (Lat/Long, Day Of Year), which adds information related to the global environmental context.

As such, Presto was trained using a diverse set of global web-scraped datasets (Sentinel-1, Sentinel-2, Landsat, and ERA5 [47]) and produced results comparable in classification performance to convolutional networks that are orders of magnitude larger in training datasets while also requiring 10 times less computational expense to run, making it an ideal processing model for onboard satellite platforms.

The Presto framework is illustrated in Fig. 9, where multi-sensor satellite observations are converted into pixel-level time-series tokens. As shown in Fig. 9, the model applies masked sensing modeling and incorporates geospatial metadata to learn temporal representations.

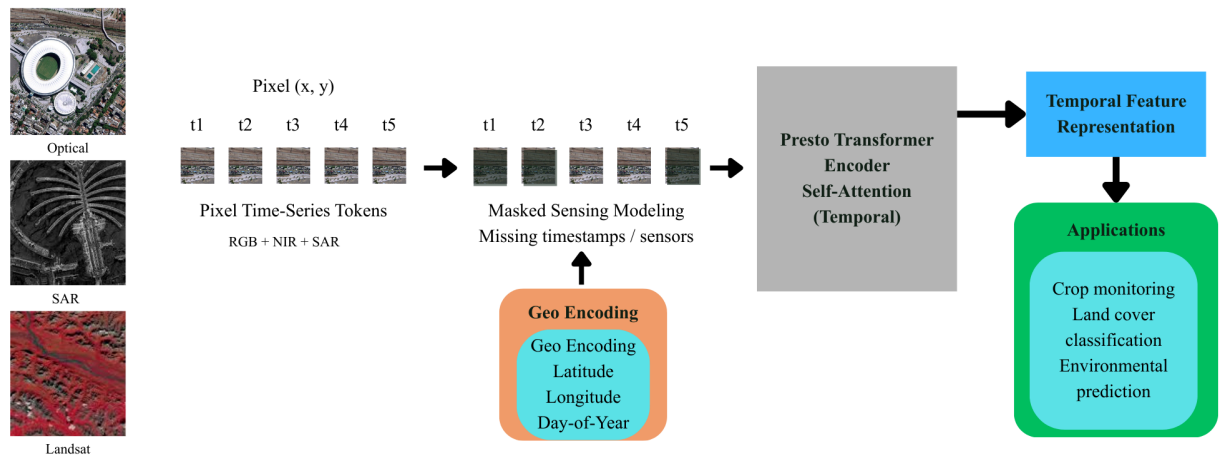


Fig. 9. Overview of the Presto framework. Multi-sensor satellite observations are converted into pixel-level time-series tokens representing the same spatial location across time. The model applies masked sensing modeling and incorporates geospatial metadata through geo-encoding. A transformer processes the temporal sequence to learn representations for downstream Earth observation tasks such as crop monitoring and environmental prediction. (Adapted from: [46])

4. Satlas (SatlasPretrain) [48]

This paper presents a framework for large-scale supervision with the use of the Swin-v2-Large [49] (197M parameters) or ResNet-50 [50] backbones.

Unlike self-supervised learning (SSL), the Satlas project will utilize multi-tasking weakly-supervised learning for their model’s multiple category training, which will include 137 different category labels (i.e., road type, solar farm, and forest cover) from open maps such as OpenStreetMap.

The SatlasPretrain dataset (302M labels, 17M images) showed that massive amounts of weakly supervised data produced better layer output performance than self-supervised methods (for example, MAE [4]) on objects where their class is already defined.

The SatlasPretrain framework is illustrated in Fig. 10, where large-scale satellite imagery is paired with weak supervision derived from OpenStreetMap annotations. As shown in Fig. 10, a vision backbone is trained using multi-task prediction heads to learn generalized geospatial representations.

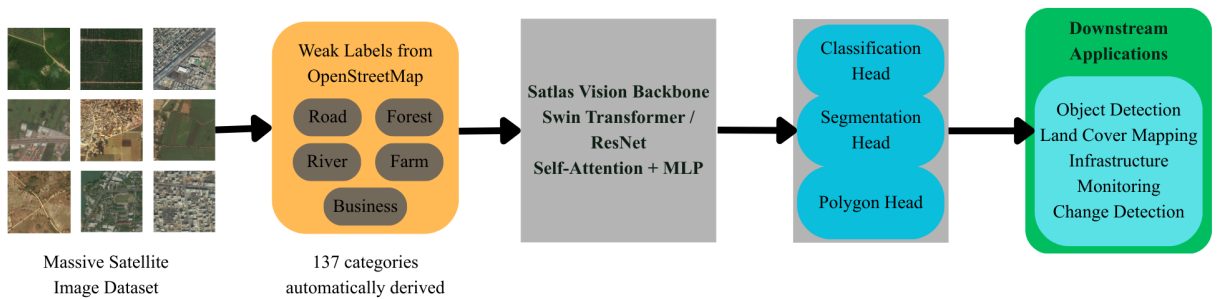


Fig. 10. Overview of the SatlasPretrain framework. A large-scale collection of satellite imagery is paired with weak supervision derived from OpenStreetMap annotations. A vision backbone such as a Swin Transformer or ResNet is trained with multi-task prediction heads to learn geospatial representations that generalize across downstream tasks. (Adapted from: [48])

5. SatVision-TOA (All-Sky Foundation Model) [51]

Global, the scale model with a very large number of parameters. The weight of the model is 632 million and, as a result, has very good performance for large-area imaging.

The images used for pre-training the model consisted of Top of Atmosphere (TOA) reflectance (radiance) images and were "masked" using a cloud-masking procedure (i.e., no clouds were present).

Supports multiple classification and regression tasks and is a foundational model for climatological and meteorological research. Training data consisted of MODIS Level 1B [52] TOA images with 14 bands, resulting in atmospheric and surface properties being treated as the signal and clouds and aerosols being treated as the noise.

The SatVision-TOA framework is illustrated in Fig. 11, where multispectral Top-of-Atmosphere (TOA) radiance data are processed using masked image modeling. As shown in Fig. 11, the model learns atmospheric and surface representations from all-sky satellite observations.

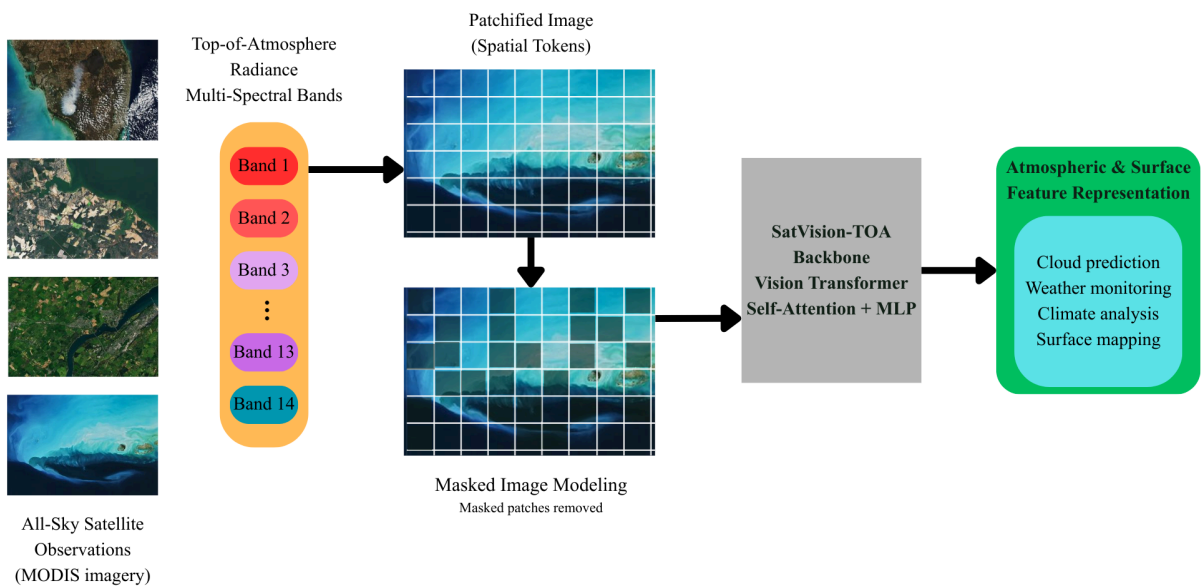


Fig. 11, Overview of the SatVision-TOA framework. All-sky satellite observations containing multispectral Top-of-Atmosphere (TOA) radiance bands are converted into

spatial tokens and trained using masked image modeling. A Vision Transformer backbone learns atmospheric and surface representations that can be used for downstream Earth observation tasks such as cloud analysis, weather monitoring, and climate studies. (Adapted from: [51])

6. Prithvi-EO-2.0 (Versatile Multitemporal FM) [5]

The Prithvi-EO-2.0 is based on a scalable ViT [7] backbone with two parameterized versions of 300 million (ViT-Large) and 600 million (ViT-Huge). The model has 3D patch embeddings to process spatiotemporal cubes (time x height x width) and also includes explicitly injected location (lat/lon) and temporal (year/day) signals into the embedding space through metadata encoders.

Prithvi-EO-2.0 has employed Spatiotemporal Masked Autoencoding (MAE) [4] with a 75% masking ratio, in which the model is required to reconstruct the missing pixels of a video sequence of 4 timestamps. Additionally, a unique 'Metadata Drop' mechanism is applied to the training process to enforce the model to learn robust features that can generalize well to new data, even with no geospatial metadata available.

The model has been pre-trained on 4.2 million Harmonized Landsat Sentinel (HLS) [61] time series chips and shows an 8% increase in performance (quantified using the GEO-Bench framework [62]) compared to its previous model. Additionally, the model has demonstrated an excellent multimodal capability by fusing together optical data, SAR data (Sentinel-1), and weather (MERRA-2 [63]) for complex downstream tasks such as biomass estimation.

The Prithvi-EO-2.0 framework is illustrated in Fig. 12, where multitemporal satellite imagery is organized into spatio-temporal data cubes and processed using a transformer-based architecture. As shown in Fig. 12, masked modeling enables the model to learn representations across both spatial and temporal dimensions.

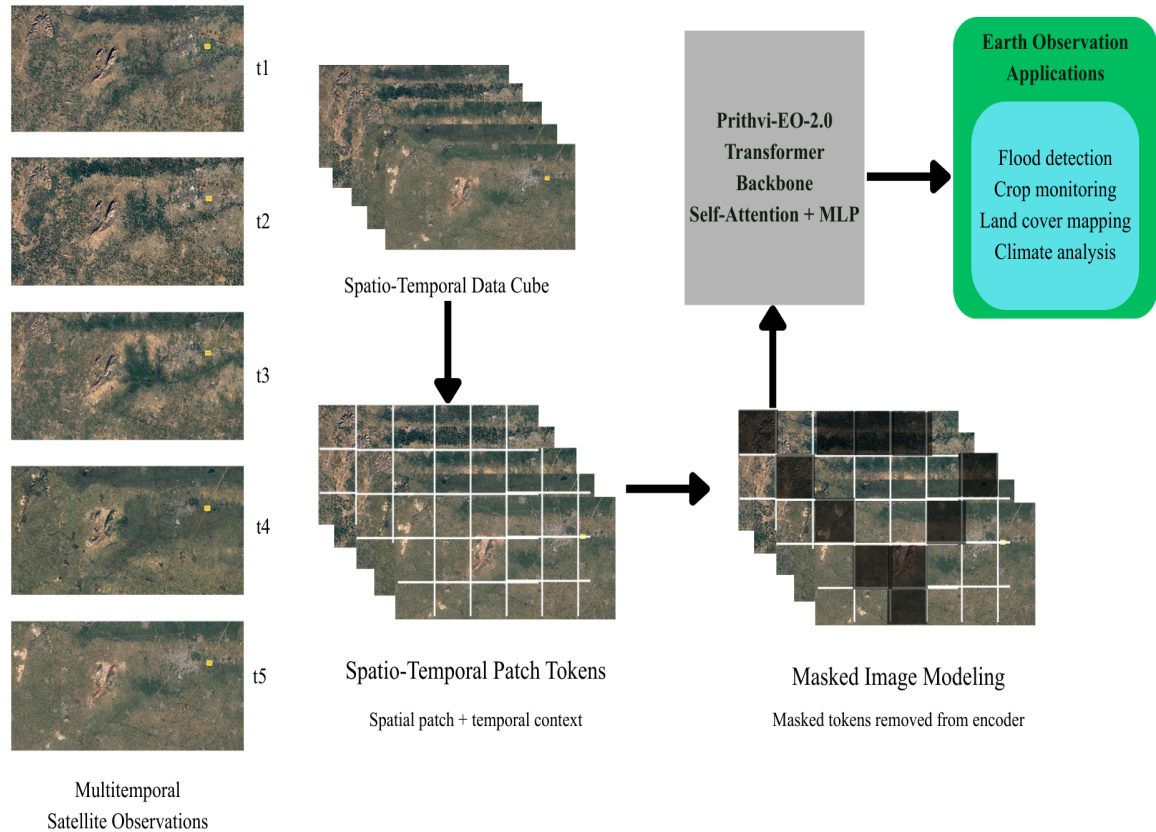


Fig. 12. Overview of the Prithvi-EO-2.0 framework. Multitemporal satellite imagery is organized into spatio-temporal data cubes and converted into patch tokens. The model is trained using masked image modeling with a transformer backbone to learn representations for downstream Earth observation tasks such as flood detection, crop monitoring, and land cover mapping. (Adapted from: [5])

7. SatMamba (State Space Models for RS) [53]

SatMamba utilizes a Mamba [54] (state space model) backbone instead of transformer attention mechanisms to achieve a linear computation complexity of $O(n)$.

This system is based on the Masked Image Modeling (MIM) methodology used for non-transformatory designs.

SatMamba is capable of addressing the Very High Resolution (VHR) images with huge token counts. For instance, if you take the following, 4096 x 4096 pixels, and compute with $O(n^2)$, it will take far longer to process than if you compute using $O(n)$. As such, this represents much more efficient computing at the same level of processing.

The SatMamba architecture is illustrated in Fig. 13, where satellite image tokens are processed using a state-space model instead of traditional self-attention mechanisms. As shown in Fig. 13, the model efficiently captures long-range dependencies for remote sensing tasks.

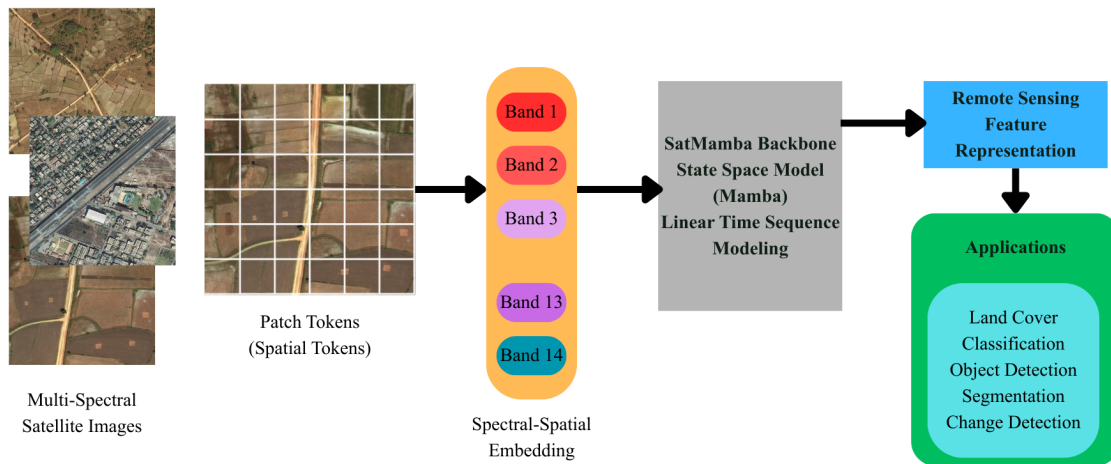


Fig. 13. Overview of the SatMamba framework. Multi-spectral satellite imagery is converted into spatial patch tokens and embedded with spectral information. The tokens are processed by a Mamba state-space backbone that efficiently models long-range dependencies. The learned representations are used for downstream Earth observation tasks such as land cover classification, segmentation, and change detection. (Adapted from: [53])

8. AnySat (Universal Modality Model) [55]

AnySat is a unified model designed around a scalable Joint Embedding Predictive Architecture (JEPA) [56]. It employs Scale Adaptive Spatial Encoders to utilize data from any sensor type as input.

A self-supervised JEPA (as opposed to a reconstructing-pixels, i.e., MAE) model is created by predicting the latent representations of masked-out regions of an image dataset instead of reconstructing the pixelated image data, which focuses on a more semantic approach rather than a computationally expensive approach to improve the efficiency of the model.

Trained on GeoPlex, it contains data collected from 11 different sensor types (LiDAR and hyperspectral sensors, among others). Its "superpower" is Zero-Shot Modality Transfer (ZSMT), which enables it to process and understand new sensor types without having been explicitly trained on them.

The AnySat framework is illustrated in Fig. 14, where multi-modal satellite data, including optical, SAR, and hyperspectral inputs, are converted into a unified token representation. As shown in Fig. 14, the model learns a shared embedding space for cross-modal understanding.

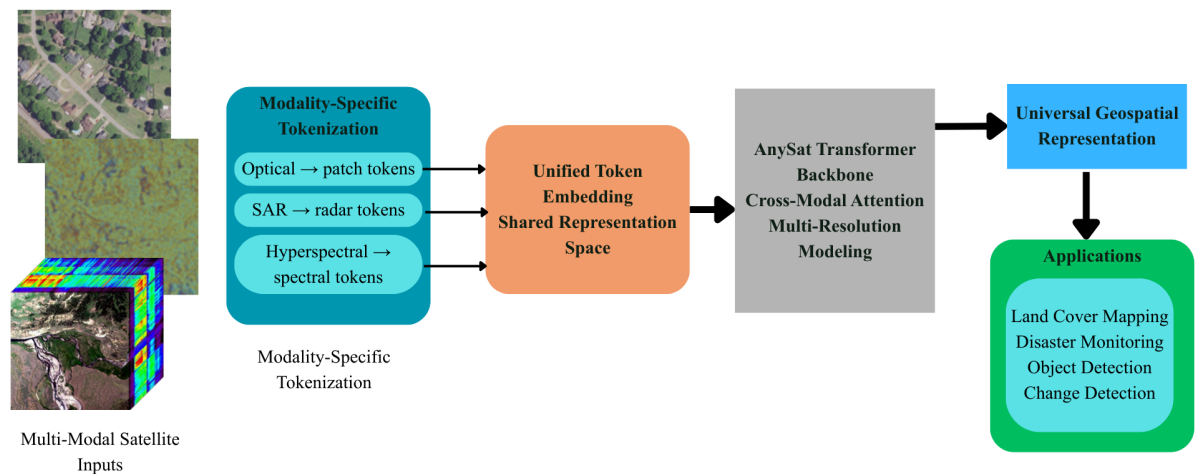


Fig. 14. Overview of the AnySat framework. Multi-modal satellite observations, including optical, SAR, and hyperspectral imagery, are converted into modality-specific tokens. These tokens are projected into a unified embedding space and processed by a transformer backbone to learn universal geospatial representations that generalize across multiple Earth observation tasks. (Adapted from: [55])

9. SpectralEarth (Hyperspectral Specialist) [57]

A unique VIT-B (Transformer Version) has been developed for processing multidimensional data, specifically for hyperspectral (hundreds of contiguous bands) spectral texture. The method used to train the VIT-B is based on multi-dimensional spectral MAE (mean absolute error) and is designed for spectral reconstruction purposes rather than just texture reconstruction.

The VIT-B uses data from the SpectralEarth Dataset [58] (a multitemporal hyperspectral database of imagery) to train and develop the VIT-B's ability to reconstruct spectral textures, providing evidence of the "curse of dimensionality" issue that currently limits the ability to use FMs trained on typical multispectral products (e.g., Sentinel 2) to perform well on hyperspectral tasks (i.e., FMs trained on spectral-trained models performed with a relative accuracy of 93.5% versus 62.6%).

The SpectralEarth framework is illustrated in Fig. 15, where hyperspectral satellite data is decomposed into spectral bands and processed as spectral tokens. As shown in Fig. 15, the model learns representations based on spectral signatures across wavelengths.

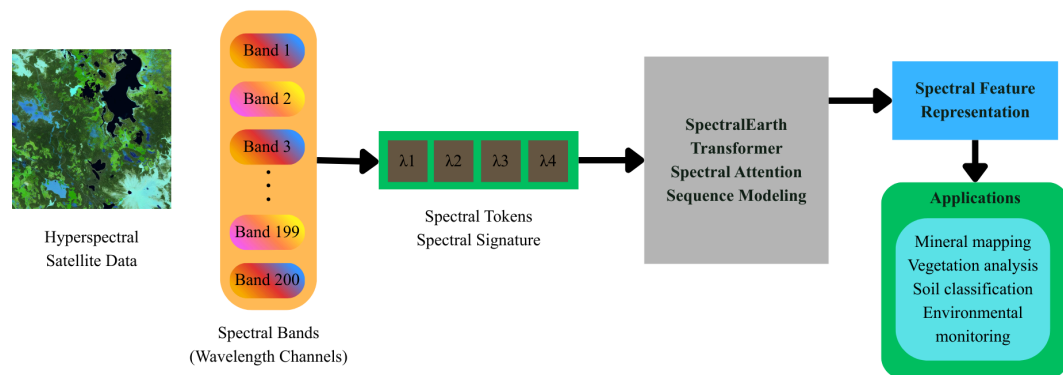


Fig. 15. Overview of the SpectralEarth framework. Hyperspectral satellite observations are decomposed into spectral bands and converted into spectral tokens representing wavelength signatures. A transformer-based backbone learns spectral representations that

support downstream Earth observation tasks such as mineral mapping, vegetation analysis, and environmental monitoring. (Adapted from: [58])

10. TerraMind (Generative Multimodal FM) [59]

The architecture of the model consists of a decoder-only architecture (like the GPT-3 family [60]) with 600 million parameters compared to the encoder-only architecture of the ViT [7]. The model was trained using next-token prediction and correlation learning. The model learns to predict the optical modality of a given image based on the synthetic aperture radar (SAR) modality of the same image.

400 million multimodal pairs were used to train the model. The model can perform cross-modal generation (i.e., synthesizing a realistic optical image from a radar input so that one could see through clouds). The model can also generate digital elevation models (DEMs) from flat, two-dimensional images.

The TerraMind framework is illustrated in Fig. 16, where multiple modalities, including satellite imagery, textual data, geospatial coordinates, and temporal information, are fused into a unified representation. As shown in Fig. 16, the model enables multimodal reasoning for Earth observation tasks.

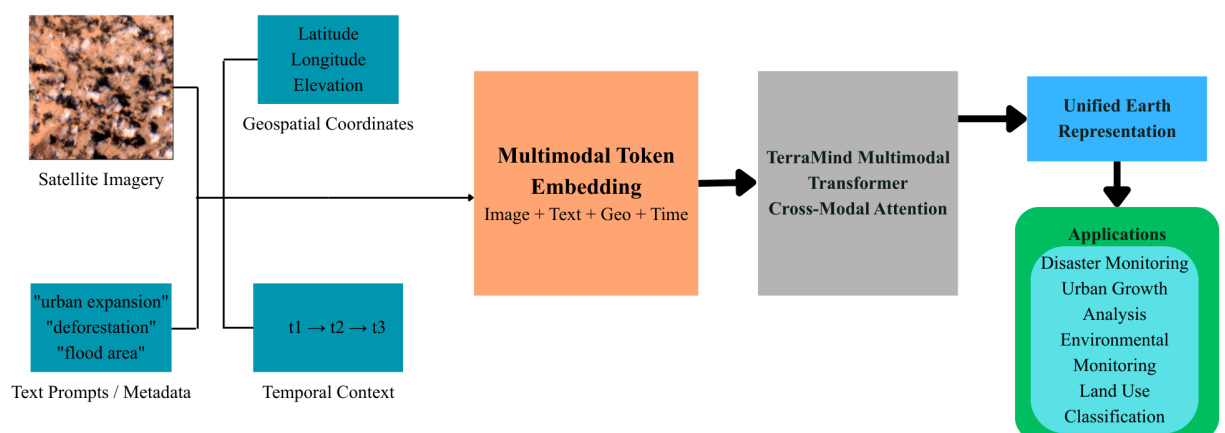


Fig. 16. Overview of the TerraMind framework. Multiple Earth observation modalities, including satellite imagery, textual metadata, geospatial coordinates, and temporal information, are converted into multimodal tokens and processed by a transformer-based

backbone. The model learns unified geospatial representations that support downstream reasoning tasks such as environmental monitoring, urban analysis, and disaster assessment. (Adapted from: [59])

TABLE 10. Comparative Summary of Key Remote Sensing Foundation Models

Model	Backbone Arch.	Parameters	Pre-training Objective	Data Scale / Source	Key Capability
SatMAE	ViT-Large	304M	Temporal + Spectral MAE	fMoW-Sentinel	Improved spectral disentanglement via group-wise embeddings.
Scale-MAE	ViT-Base	86M	Scale-Aware MAE	fMoW + Million-AID	GSD-conditional encoding for resolution independence.
Satlas	Swin-v2-L	197M	Multi-task Weak Supervision	SatlasPretrain (≈ 302 M labels)	Strong semantic representations from massive weak supervision.
Presto	Transformer	~ 20 M	Masked Sensor Modeling	Global Satellite Time-Series	Lightweight, efficient pixel-level temporal analysis.
SatVision-TOA	ViT-Huge	632M	All-Sky MAE	MODIS L1B (TOA Radiance)	Atmospheric and cloud-aware representation learning.
Prithvi-EO-2.0	ViT (Scalable)	300M / 600M	Spatiotemporal MAE	1.48B tokens (HLS + SAR)	Unified multi-sensor and metadata-aware representations.

SatMamba	Mamba (SSM)	Scalable	Self-Supervised Sequence	Standard RS Archives	Linear-complexity ($O(N)$) modeling for very large scenes.
AnySat	JEPA (ViT)	Scalable	Joint Embedding Prediction	GeoPlex (11 Sensors)	Sensor-agnostic generalization and zero-shot transfer.
SpectralEarth	ViT-Base	86M	Spectral MAE	SpectralEarth (HSI)	High-dimensional hyperspectral representation learning.
TerraMind	Gen. Transformer	~600M	Next-Token Multimodal Pred.	~400M Multimodal Pairs	Cross-modal generation and modality completion.

6. Case Studies on Road Extraction & Cloud/Cloud Shadow Segmentation

6.1. Road Detection and Extraction Using Foundation Models

1. Problem Definition and Structural Challenges

The term "road detection and extraction in remote sensing" refers to the processing and identification of road components from the remotely sensed images, considering each pixel individually. Importantly, unlike the classes in other semantic segmentation tasks, which tend to occupy localized and contiguous spatial locations, in the structured prediction task of road extraction, the foreground is composed of thin, elongated, and highly connected linear elements. Roads stretch the width of an image, which means that they form complex graph-like objects of various intersections, branches, and varying curvature. Therefore, the prediction is not just pixel-wise classification but also

necessarily incorporates considerations for the spatial organization of linear objects [82] [83].

Structural characteristics of roads in satellite imagery make this task uniquely challenging:

- 1) High Aspect Ratio and Thin Geometry

Roads typically occupy a small number of pixels in width while extending over long distances. In high-resolution imagery, minor localization errors can completely disconnect road segments, leading to topological inconsistencies. This sensitivity to boundary precision distinguishes road extraction from area-based segmentation tasks such as building or vegetation mapping [84].

- 2) Topological Continuity Requirement

Correct road extraction is not solely defined by pixel accuracy but by the preservation of network connectivity. Disconnected predictions, even if locally accurate, significantly reduce the usability of extracted road maps for routing and navigation applications. This introduces a structural constraint that is not explicitly enforced by conventional loss functions such as cross-entropy or IoU [85].

- 3) Extreme Class Imbalance

Road pixels typically constitute a very small fraction of total image pixels, especially in rural or semi-urban scenes. This imbalance biases learning toward background classes and makes optimization unstable, particularly for models trained with standard pixel-wise objectives [86].

2. Importance and Real-World Applications

The road network is one of the most fundamental layers in terms of geospatial information, constituting the lifeline of human mobility, economic activity, and emergency response. Hence, the importance of road maps with accurate and up-to-date content, deriving from remote sensing imagery, constitutes a need for various applications ranging from civilian to governmental and humanitarian use and is also beneficial for informed decision-making in time-critical and resource-constrained scenarios.

In geospatial analysis, roads are often used as a structural prior for many of the more advanced reasoning tasks. Transportation networks define the concept of accessibility, connectivity, and relationships between two regions in space. This means that when road data is erroneous, other downstream methods and applications will be adversely affected, including routing, service access analysis, population mobility models, etc. [87]. Traditional map databases often lag behind real-world changes due to the cost and latency of manual annotation. Satellite-based road extraction offers a scalable alternative, capable of providing timely updates over large geographic extents where ground surveys are infeasible.

Urban Planning and Infrastructure Monitoring [88]

In urban planning and infrastructure monitoring, patterns of urban growth, transportation efficiency, and infrastructure accessibility are analyzed from extracted roads. Satellite imagery is used to derive road density and connectivity that helps planners identify underserved regions, monitor congestion patterns, and guide development strategies. Foundation-model-based road extraction is important in regions where roads vary widely in width, material, and layout. Such models reduce the need for retraining and manual intervention when applied to different geographic contexts.

Disaster Response and Emergency Management [89]

Roads become an important factor for rescue and relief operations during natural disasters. Rapid identification of blocked, damaged roads helps emergency agencies to plan evacuation routes, provide resources, and reach the affected regions. Foundational

models pre-trained on diverse environmental conditions, provides improved resilience even for post-disaster road extraction, which is a major challenge due to debris, shadows, water coverage, and atmospheric interference.

Humanitarian Mapping and Low-Resource Regions [90]

In many conflict-affected regions, reliable road maps are majorly incomplete or entirely absent. Organizations rely on baseline maps from satellite imagery to provide support in healthcare, food, vaccination, and evacuation. Automated road extraction significantly reduces the dependency on manual mapping. The few-shot adaptability of foundation models is especially critical in such settings, where labeled training data is scarce or unavailable, yet rapid deployment is essential.

Navigation, Logistics, and Autonomous Systems [85]

Accurate network extraction also helps develop modern navigation systems, logistics optimization, and autonomous platforms. Satellite-derived road maps, along with ground-based sensing, provide a global, consistent view, particularly in regions where GPS traces or street-level imagery is limited. Preserving topological correctness, such as intersections, merges, and connectivity, is essential for these applications. Fragmented or disconnected road prediction can lead to major failures, demanding a need for models capable of overall structural reasoning rather than local classification.

3. Classical Approach

Prior to the mass adoption of deep learning, road detection and extraction from satellite imagery was mostly done using classical computer vision and machine learning techniques. These approaches faced constraints like limited spatial resolution, limited spectral information, and high variations.

Edge and line based methods [91] [82]

Earlier, road extraction techniques focused on edges and linear structures, majorly since roads appear as elongated regions with intensity different to their surroundings. Common approaches included the Canny or Sobel operators as major edge detectors. These

methods were often combined with directional filtering and thresholding to suppress noise and highlight linear features. However, the performance was very sensitive to illumination changes, shadows, and background clutter. False positives were produced frequently for urban scenes, building edges, and field boundaries, while entire road segments were missed due to occlusions and low contrast.

Morphological and region-growing techniques [92]

Morphological operations such as dilation, erosion, opening, and closing were widely applied to address fragmentation and noise in edge based outputs. Region-growing algorithms further attempted to detect complete road segments based on spectral similarity and spatial continuity. While these methods improved connectivity, they relied on fixed structuring elements and manually tuned parameters, which varied significantly across resolutions and sensors. Therefore, these methods lacked robustness and required extensive human intervention.

Graph-Based and Network Optimization Approaches [93]

Since roads are by their very nature network-like, a number of methods have formulated road extraction as a graph inference problem. Road segments or pixels that were detected were shown as nodes, with edges signifying directional consistency or spatial proximity. Global connectivity constraints were then enforced using graph optimization techniques, minimum spanning trees, and shortest-path algorithms. Topological consistency was enhanced by graph-based models, but their effectiveness was largely reliant on the caliber of initial detections. Early-stage errors, like noisy or missing segments, spread throughout the graph and reduced overall performance. Furthermore, scaling these techniques to large satellite scenes was challenging and computationally costly.

Feature-Based Machine Learning Methods

In order to separate road pixels from background, classifiers like Support Vector Machines (SVMs) and Random Forests were combined with handcrafted feature descriptors, such as texture measures, spectral indices, gradient histograms, and shape descriptors, as traditional machine learning gained popularity. These methods allowed for

data-driven decision boundaries, which was an improvement over strictly rule-based systems. However, feature design was closely linked to their success. Additionally, handcrafted features had limited generalization across different landscapes and imaging conditions because they were unable to capture long-range dependencies and complex spatial context.

4. CNN Based-road extraction Methods

The introduction of CNN marked a significant shift into satellite based road detection and extraction. By enabling end-to-end learning directly from data, CNNs removed many of the limitations associated with classical methods.

By substituting end-to-end semantic segmentation for manually constructed feature pipelines, Convolutional Neural Networks (CNNs) created the first reliable data-driven framework for satellite-based road detection and extraction. Dense pixel-level prediction was made possible by encoder–decoder architectures like U-Net [18], which preserved the fine spatial details necessary for defining thin road structures. Dilated convolutions and multi-scale context aggregation were added to later models, such as DeepLab [94] and D-LinkNet [95], to enhance continuity and performance on satellite benchmarks like DeepGlobe [96].

CNN-based methods are still intrinsically limited by the locality of convolutional operations, despite these advancements. It is necessary to infer global road connectivity indirectly using stacked receptive fields, which frequently leads to fragmented predictions in situations with low contrast, shadowing, or occlusion, all of which are frequent in satellite imagery. Additionally, CNN models usually require extensive fine-tuning or retraining for new geographic domains due to their limited cross-sensor and cross-region generalization and fully supervised training.

Although CNNs greatly improved satellite road extraction over classical techniques, scalability and structural reasoning at large spatial extents are limited by their architectural bias toward local feature aggregation and task-specific supervision. The

shift to foundation-model-based techniques that can explicitly model global context and have better transferability is driven by these limitations.

5. Transition to Foundation Models

Transformer-based foundation models have gained popularity due to the structural drawbacks of CNN-based road extraction, specifically locality bias and limited cross-domain generalization. In contrast to convolutional architectures, Vision Transformers (ViTs) [3] use self-attention mechanisms to directly model long-range spatial dependencies. This is important for maintaining road continuity in large satellite scenes.

In parallel, large-scale self-supervised pretraining approaches like Masked Autoencoders (MAE) [4] have shown that good image representations can be learned on very large amounts of unlabeled satellite imagery. More recent foundation models like SatMAE [17] and Prithvi [5] extend this to multispectral and geospatial imagery. In this sense, the shift to foundation models can be seen as a move away from task-supervised learning and towards more general geospatial representation learning, which matches the global structural nature of road extraction via satellite imagery.

6. Foundation Models Relevant to Road Extraction

A. SatMAE [17]

SatMAE is an extension of the masked autoencoder concept introduced in the Masked Autoencoders Are Scalable Vision Learners paper, but it is designed to work with satellite imagery using a Vision Transformer encoder. Unlike the traditional MAE models trained on RGB images, the embeddings are designed to work with satellite imagery and, in some cases, with time stacks of the imagery. In the pretraining phase, a high percentage of the spatial patches are randomly masked, and the transformer encoder only processes the visible tokens, with a lightweight decoder being used to reconstruct the masked information. This allows the encoder to learn informative and compact geospatial features from large volumes of unlabeled satellite imagery.

For road extraction, the architecture of SatMAE is especially well-suited due to the ability to leverage both spectral and long-range spatial dependencies. Roads have different spectral characteristics compared to the soil or other built-up areas and may be partially covered by vegetation or shadows. The goal of masked reconstruction forces the encoder to reason about contextual continuity, which is beneficial for segmentation tasks where the object of interest has a small height and a long length. Evaluations of the proposed approach by the original authors, “SatMAE: Pre-training Transformers for Temporal and Multi-Spectral Satellite Imagery,” NeurIPS Workshop, 2022, show improved transfer results on segmentation benchmarks compared to the ImageNet pre-trained baselines.

The SatMAE framework is illustrated in Fig. 7, where masked image modeling is used to learn robust spatial representations from satellite imagery. As shown in Fig. 7, the reconstruction of masked regions helps the model capture structural continuity and contextual relationships, which are essential for accurate road extraction in heterogeneous terrains.

B. Prithvi [5]

Prithvi, which is a collaborative effort between IBM and NASA, can be considered a large-scale geospatial foundation model that uses Harmonized Landsat-Sentinel (HLS) images for pretraining. The model's architecture consists of a Vision Transformer backbone with masked autoencoding objectives, which can be extended to accommodate multispectral and temporal stacks of images. The model utilizes large-scale Earth observation datasets to learn domain-specific representations that include seasonality, illumination, and geographic differences. The model's transformer-based encoder can process spatial patches with geospatial positional embeddings, which can be used to learn contextual dependencies over large spatial regions.

From the perspective of road extraction, Prithvi has strong cross-region generalization and robustness. It is necessary to delineate road networks over various terrains and climatic conditions. However, models trained on localized datasets cannot generalize well. But with large-scale pretraining, fine-tuning with less labeled data can be done with stable performance.

The Prithvi-EO-2.0 framework is illustrated in Fig. 12, where multitemporal satellite imagery is modeled as spatio-temporal data cubes. As shown in Fig. 12, the model captures temporal consistency and structural patterns across time, which helps in accurately detecting road networks despite occlusions, seasonal variations, and changes in illumination.

C. Scale-MAE [23]

The Scale-MAE model extends the application of masked autoencoders to handle the scale variations in the images. The Scale-MAE model is based on the encoder-decoder structure of the Vision Transformer. The model undertakes the task of masked patch reconstruction with the inclusion of scale-aware input sampling. Unlike the traditional MAE models, Scale-MAE takes into consideration the multi-scale features during the pretraining phase.

This is especially important for satellite-based road extraction, where roads appear as a wide textured area in high-resolution images and a thin line in medium resolution images. By learning scale-invariant geospatial embeddings, Scale-MAE achieves better transferability and structural continuity in the segmentation task. As demonstrated by the experimental results provided by “Scale-MAE: A Scale-Aware Masked Autoencoder for Remote Sensing Pretraining,” CVPR 2023, the proposed approach achieves better transferability compared to the traditional MAE and ImageNet methods.

The Scale-MAE framework is illustrated in Fig. 8, where multi-resolution satellite imagery is combined with Ground Sample Distance (GSD) information to learn

scale-aware representations. As shown in Fig. 8, this allows the model to detect road networks consistently across varying spatial resolutions, where roads may appear differently in terms of width, texture, and visibility.

D. SkySense [80]

The large-scale foundation model for remote sensing, named SkySense, has been proposed with the aim of incorporating multi-task supervision and multimodal supervision in a transformer architecture. Unlike other self-supervised learning methods based on masked modeling, the proposed model also considers weakly supervised learning objectives with diverse tasks. The model is based on a transformer architecture, which can effectively handle heterogeneous input modalities of satellite images.

In the case of road extraction, the model provides a rich spatial-semantic embedding that includes both scene layout and local feature details. The multi-task pretraining paradigm allows the model to learn the relationships within the image, which can be beneficial for connectivity and segmentation stability.

The SkySense framework is illustrated in Fig. 17, where high-resolution satellite imagery is processed using a feature extraction backbone followed by a segmentation network for pixel-wise prediction. As shown in Fig. 17, the model effectively captures fine-grained spatial details, enabling accurate extraction of road networks in complex urban and rural environments.

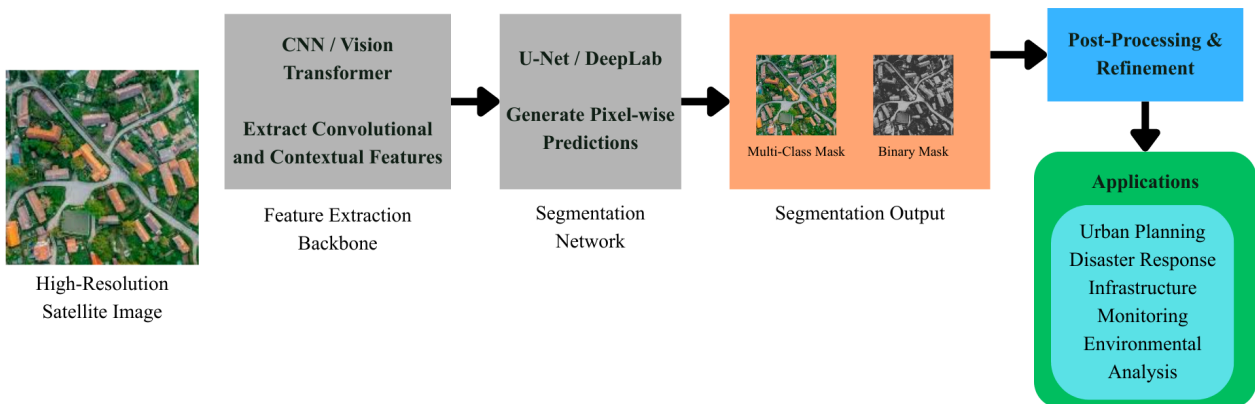


Fig. 17. Overview of the SkySense framework. High-resolution satellite imagery is processed using a deep learning backbone for feature extraction, followed by a segmentation network that generates pixel-wise predictions. The output masks can be refined using post-processing techniques to produce accurate geospatial representations for downstream applications. (Adapted from: [80])

7. Datasets and Benchmarks for Road Detection

To effectively evaluate satellite image-based road extraction techniques, there is a need to test these techniques using diverse sets of benchmark datasets with different spatial resolutions, geographic distributions, and evaluation protocols. In fact, these datasets not only evaluate the pixel-wise accuracy of these techniques but also their topological correctness and cross-region generalization.

Currently, high-resolution benchmarks such as SpaceNet and DeepGlobe remain the most widely used datasets for supervised road segmentation tasks. Medium-resolution datasets such as Sentinel-2 Roads test the performance of models on a medium spatial scale. Region-specific datasets such as WHU-Road and CHN6-CUG emphasize the importance of structural variation in complex urban and rural settings. In addition, graph-based datasets such as RoadTracer emphasize the importance of topological preservation. Table 11 summarizes major datasets used for satellite-based road detection, highlighting variations in spatial resolution, annotation quality, and geographic diversity, which significantly influence model performance.

TABLE 11. Major Datasets for Satellite-Based Road Detection

Dataset	Year	Resoluti on	Geographic Scope	Annotation Type	Evaluation Focus	Key Challenges
Massachusetts Roads	2013	1.0 m	Regional (USA)	Rasterized centerlines (OSM)	IoU, Precision/Rec all	Label noise (width), varying context, occlusion.

SpaceNet 3 / 5	2018	0.3 m	Urban (4 Cities: Vegas, Paris, etc.)	Graph (Centerlines)	APLS, TOPO-IoU	Topology preservation, complex intersections.
DeepGlobe Roads	2018	0.5 m	Global (India, Indonesia, Thailand)	Pixel-level binary masks	IoU, F1-Score	Surface variability (paved/unpaved), rural connectivity.
RoadTracer	2018	0.6 m	Urban (40 Cities)	Graph (OSM Centerlines)	Junction Matching, Connectivity	Graph extraction, dense urban morphology.
WHU-Road	2019	0.3–1.0 m	Regional (China)	Manual pixel-level masks	IoU, F1-Score	Shadow occlusion, complex texture similarity.
CHN6-CUG	2021	0.5 m	Urban (6 Cities, China)	Manual masks	IoU, Completeness	Generalization across diverse urban densities.
Sentinel-2 Roads	Varies	10 m	Regional / Global	Weak / Manual labels	IoU (Coarse)	Low resolution, mixed spectral pixels, discontinuity.

These datasets collectively provide complementary evaluation settings across resolution scales, geographic regions, and structural complexity, enabling systematic assessment of both task-specific and foundation-model-based road extraction approaches.

6.2. Cloud vs Cloud-Shadow Segmentation

1. Problem Definition

Identifying cloud vs. cloud shadow segmentation is one of the fundamental challenges in multispectral satellite imagery. It is the pixel level classification. Classifying images into 4 distinct classes that are clear sky, thin cloud, shadow, and semi-transparent cloud. The problem lies with the spectral characteristics of the cloud. Clouds generally have a high reflectance across all the bands, which causes confusion between bright objects like snow, ice, bright rooftops, etc. The other problem lies with the cloud shadows; these shadows are identical to water bodies, mountains, or dark soil. Identifying these tasks is complicated due to spectral confusion. The goal is to precisely mask these cloud shadows and clouds for segmentation.

2. Real-world applications

Precision Agriculture & Food Security:

Clouds and shadows are the major challenges of automated crop monitoring. In precision agriculture, farmers are dependent on satellite data to calculate indices like the Normalized Difference Vegetation Index (NDVI), which is used to measure plant health. The clouds are bright and highly reflective where the shadows are darker. When a cloud shadow falls on a crop, the satellite images misinterpret the dark pixels as water, soil, or any other vegetation. Whereas thin cirrus clouds can brighten up the images, leading to the wrong calculations. Cloud segmentation algorithms help create cloud masks that mark unusable pixels. Which helps in accurate yield forecasting, automated irrigation, and pest and disease monitoring.

Disaster Response & Management:

When a disaster is hit, during rapid response scenarios, satellite imagery is the first tool used to look for damages, but weather events such as hurricanes and typhoons have large and big clouds. Water bodies absorb light, which appears dark in satellite images, and cloud shadows also appear dark. A scenario where a system may classify cloud shadow as a water body or floodwater, leading to wrong estimation.

Segmentation of cloud shadows can help rescuers define boundaries of floodwaters. Ensuring rescue boats reach the right place. Wildfire segmentation helps distinguish between white clouds and grey / brown smoke plumes.

Climate Change & Carbon Accounting:

Measuring deforestation and glacier melt, scientists built mosaics from different days. If clouds are not perfectly segmented, they degrade the mosaics.

Major applications include carbon verification and energy budgets. Carbon verification describes the process of trading carbon credits based on preserving forests between countries and corporations. It allows for automated monitoring of forest cover protection. Earth's energy budget plays a vital role in climate change, as it reflects light but traps heat.

Solar Energy Forecasting:

Real-time monitoring of electricity by grid operators is important to balance electric supply. A cloud passing by may cause a sudden drop in power generation in a solar farm. These sudden drops can destabilize the power grids if backup generators are not available. Major applications include solar "nowcasting" and grid stability. Solar nowcasting is the process of tracking cloud motions by segmenting clouds based on satellite cameras. They can predict how long a cloud will shade a farm. Grid stability ensures the operator turns up gas turbines or backup battery storage for 10-15 minutes prior to burnouts.

Urban Planning & Infrastructure:

Urban planners use satellite data to monitor rapidly changing cities. Urban cities are a complex mixture of concrete water bodies and green parks. Bright concrete is usually confused as small clouds, while water bodies can be detected as cloud shadows. The application includes surface mapping and urban heat hotspots. Surface mapping of concrete coverage area, while not mistaken as a cloud, is important to manage and plan urban infrastructure. Urban heat hotspots ensure that the measured land surface temperature is valid and not obstructed by clouds or cloud shadows.

3. Classical Approach and CNN

The classical method for cloud and cloud shadow detection uses a thresholding approach, which is simply rigid if the rules are then applied to individual pixels. The image is split into its spectral components, which isolates the blue channel to detect clouds, proving that clouds usually reflect more light than ground or cloud shadow. We hardcoded a threshold to enforce a rigid rule, i.e., “anything that is above the threshold brightness must be a cloud.” Similarly for shadows, we used a red channel and applied a low-intensity threshold, proving that where sunlight is blocked it will be registered as the darkest pixel in the image.

This illustrates a critical challenge of the classical method known as "threshold rigidity." It exists because the classical method processes isolated pixels without spatial awareness, making it difficult to differentiate between objects of similar spectral values. It eventually fails in complex environments such as misclassifying snow as cloud or dark water bodies or dense forest as shadow. These challenges forced us to transition into modern methods like machine learning algorithms and neural network based CNN architecture.

To overcome the challenges of threshold based methods, the focus shifted towards early machine learning algorithms using statistical probability. Unsupervised methods like K-means clustering group unlabelled pixels based on their spectral similarity, organizing the image effectively into clusters of bright and dark pixels. This pixel was refined by supervised methods like Support Vector Machine (SVM), utilizing labeled training data to calculate boundaries between cloud, cloud shadow, and ground classes. An ensemble method like random forest uses hundreds of decision trees, improving the accuracy and reducing noise as well as overfitting. However, similar to classical methods, these methods remained largely pixel-based, which analyzes the color of a particular point rather than understanding surrounding context.

The introduction of CNNs overcomes these pixel based limitations through spatial-aware reasoning. CNN utilized convolutions to scan the image, recognizing high-level features such as texture, shape, and edges. This allows the model to finally distinguish between

objects of identical spectral values . It solved the confusion that challenged earlier models by learning the spatial structure.

4. Foundational models

Foundational models are found to be superior to classical threshold methods and convolutional neural networks. CNN's ability to perform depends upon global spatial reasoning and self-supervised representations. Classical methods rely on rigid and defined spectral thresholds that variably fail under different atmospheric conditions due to high albedo surfaces such as snow and ice.

The foundational model solves this by removing the magnifying glass and looking at the image entirely simultaneously. Using a self-attention mechanism model is able to link a specific cloud to its distant shadow by calculating the geometric offset based on the sun's position. This allows the model to realize that the dark spot is not a random object but the presence of clouds elsewhere in the image. This is where CNN fails: where it needs to maintain a logical connection.

KEY FOUNDATIONAL MODELS FOR SEGEMENTATION

1. Prithvi-EO-2.0 (IBM & NASA)

Prithvi-EO-2.0 is trained on 600 million parameter vision transformers developed by IBM and NASA [5]. Prithvi is trained on millions of multispectral satellite images. It performs well in cloud vs. cloud shadow segmentation due to autoencoder training. It specifically teaches the model to reconstruct masks or hidden satellite data. By looking at different spectral bands along with spatial and temporal metadata. It accurately defines the clouds and cloud shadow.

Beyond just detecting and masking clouds Prithvi EO 2.0 is good at gap imputation, which allows it to fill the landscape details hidden under cloud cover [5]. This precise segmentation is inserted into the model to perform gap imputation. When processing realtime data, Prithvi is able to segment clouds and

mask them. It evaluates the cloud and unclouded edges, reconstructing hidden layers and performing well in cloud segmentation.

2. SkySense (Wuhan University & Ant Group)

SkySense is one of the powerful Multi Modal remote sensing foundational models. Unlike other models which rely on single data type, it parallelly processes high resolution optical images, multi spectral data and SAR imagery [80]. Which makes models effectively work on cloud segmentation. As it incorporates SAR data, which works on radio waves that penetrate through thick clouds, helping the model to accurately detect and map ground features even when visibility is zero. Many changes in SkySense have streamlined the technology. As the separate neural networks are replaced by transformers and by reducing the parameters, which increases the efficiency of overall performance. The model deeply learns how specific radar backscatter patterns and correlated optical features under clear conditions. When a thick cloud completely blocks the view, SkySense cross-references uninterrupted SAR data and learns multi-modal memory and accurately performs cloud segmentation.

3. SatMAE (Stanford University)

SatMAE is a masked autoencoder framework that is specially designed for complex and multidimensional satellite imagery [17]. Unlike other models, it processes both multispectral and temporal data by the method of spectral positional encodings.

It stands out from different models by independent masking. While training, it hides 75% of the image. The model reconstructs the missing visible data using infrared bands or previous images. For cloud segmentation the model is effective, as if thick clouds block the optical view, the model finds the hidden landscape by cross-referencing unobstructed infrared signatures or contextual data from the past [17].

4. Scale-MAE (University of Maryland)

It is also one of the masked autoencoders designed to solve scale blindness in earth observation [23]. Though the model struggles in understanding the physical size of objects when analyzing satellite images with varying spatial resolutions. To solve this, it uses Ground Sample Distance (GSD) Positional Encoding. Instead of locating where the pixel is, it tells us about the absolute physical area of earth, which allows the model to know real-world geometry. Additionally with this while training the model, the decoder uses a Laplacian pyramid structure to reconstruct both broad low-resolution features and sharp high-resolution edges of hidden areas [23].

5. GFM (Global Foundation Model - Spectro-Temporal)

It is a massive pre-trained model that understands planetary-scale earth observation data. It focuses on spectrotemporal processing, where models analyze critical dimensions, i.e., spectral dimension and temporal dimension. Spectrotemporal GFM utilizes advanced multidimensional vision transformers and temporal attention mechanisms, which help to learn how the surface naturally changes [97].

DATASETS FOR CLOUD VS. CLOUD SHADOW SEGMENTATION

Table 12 summarizes major datasets used for cloud and cloud-shadow segmentation, highlighting variations in spectral bands, annotation strategies, and atmospheric conditions, which significantly influence model performance.

TABLE 12. Major Datasets for Satellite-Based Cloud vs Cloud Shadow Segmentation

Dataset	Year	Resolution	Geographic Scope	Annotation Type	Evaluation Focus	Key Challenges
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38-Cloud	2018	30m (Landsat 8)	Global (17,601 patches)	Binary Ground-Truth	Baseline CNN binary segmentation	Overfitting to a single sensor; ignores shadows entirely.
SPARCS	2014	30m (Landsat 8)	Global (80 sub-scenes)	Multi-class	Differentiating snow from clouds and mapping shadow geometry.	Small dataset size severely limits model generalization.
Cloud SEN12	2022	10m (Sentinel-2)	Global (49,400 patches)	Multi-tier (Polygons, scribbles)	Benchmarking self-supervised transformer architectures.	Managing massive data volume; detecting fragmented "popcorn" clouds.
SEN12MS- CR	2020	10m (Sentinel-1 SAR & Sentinel-2)	Global (175 Regions)	Unannotated Paired Triplets	Multi-modal fusion and cloud gap imputation.	Forcing networks to learn SAR-to-optical physical relationships.

7. DOWNSTREAM APPLICATIONS AND PERFORMANCE

1. Scene-Level Semantic Understanding

Task Relevance and Model Property:

The Land Use and Land Cover (LULC) is one of the first places where Foundation Models will be tested for semantic consistency. Since LULC requires the classification of global categories (like how to distinguish "Continuous Urban Fabric" from "Industrial Units") from local textures, it is very different from dense prediction tasks that typically require fine detail in the delineation of boundaries.

The Property Stress test family is specifically used to test for Spectral-Semantic Invariance. For example, a forest will appear bright green in the summer and brown in the winter but should be mapped to the same class label regardless of the variability in spectral signature. The goal of this test is to assess whether the global feature space that has been learned by the pre-training method (for example, masked autoencoding vs. contrastive learning) is disentangled and robust against seasonal and atmospheric variation.

Benchmarking Analysis:

We use the m-BigEarthNet [20] benchmark (Sentinel-2) to evaluate our performance using the multi-label classification task of identifying several concurrently existing categories (e.g., “River” + “Forest” + “Road”) within one image, penalizing models that only learn dominant features. Table 13 presents a comparative evaluation of foundation models on the m-BigEarthNet benchmark, highlighting how different pre-training strategies and architectural designs influence semantic understanding in multi-label land cover classification.

TABLE 13. Comparative Performance on Scene-Level Semantic Understanding (m-BigEarthNet) Metric: Micro-Averaged F1-Score (Higher is better)

Model	Architecture	Pre-training Paradigm	F1-Score (%)
Prithvi-EO-2.0-TL	ViT-Huge (600M)	Spatiotemporal MAE	70.93
Satlas	Swin-v2 (100M)	Multi-Task Supervised	69.63
DOFA	ViT-Large (300M)	Frequency-Adaptive MAE	68.58

DeCUR	ResNet-50 (25M)	Multimodal Contrastive	67.54
DINO	ResNet-50 (25M)	Knowledge Distillation	67.37
Scale-MAE	ViT-Base (86M)	Scale-Aware MAE	62.80

Three major conclusions about benchmarking on m-BigEarthNet:

Temporal Context as a Semantic Multiplier - The top-scoring model of Prithvi-EO-2.0 (70.93%) has demonstrated that scene understanding is fundamentally a time-based problem. Models which incorporate time in their architectures (e.g., using either 3D tokens or timestamps) provide the best means of distinguishing between vegetation classes that have different phenological cycles (e.g., arable land vs. pasture). In contrast, models that have only one snapshot of a given site (such as DOFA) are hampered by their inability to distinguish between classifications due to multiple scenes having very similar spectral profiles.

Weak Supervision vs. Self-Supervision - Despite being mostly smaller (100M parameters vs. 300M) than the larger self-supervised models, Satlas (69.63%) outperforms them all. This suggests that for the purpose of providing higher-order semantic classification, the use of Massive Weak Supervision (using OpenStreetMap labels that include error) provides a stronger direct signal for separating types/classifications of objects than using pixel reconstruction objective functions (e.g., MAE) that are focused on capturing the texture rather than the semantic content of an object.

The Trade-off for "Scale-Invariance" - Scale-MAE provides very poor performance results on this fixed-resolution task (62.80%). This indicates a very clear trade-off in the design of an architecture that is optimized for achieving resolution invariance (handling 0.1m - 30m GSD) such that the added architectural costs to achieve resolution invariance further dilute the ability of the model to provide spectral discrimination when utilizing a single resolution benchmark.

2. Dense Pixel-Level Prediction

Task Relevance and Model Property:

In order to categorize every component pixel in a pixel-level dense prediction task, remote sensing has presented the highest level of granularity (or detail) required for completion. An example of this would be characterizing all parts of a scene related to flooding or fire mapping. The high degree of specificity required to properly characterize a scene provides great difficulty when analyzing tasks that have a high degree of spatial precision and boundary sensitivity. In the case of assessing a scene through scene classification (i.e., determining the presence of a “water” scene), having an accurate classification of a “water” scene is usually considered sufficient. However, when performing segmentation to assess flooding, the model must accurately delineate the jagged edge between the flooding and the adjacent dry areas. This will demonstrate the criticality of having the model's pre-trained distance model performing well (typically trained on coarse granularity in patch size [5]) to allow for the fine-grain/high-frequency properties from the remotely sensed data for the operational mapping applications.

Benchmarking Analysis:

To account for this task and its associated characteristics, we will study the model's capability to use two types of disaster-related datasets: Sen1Floods11 [66] (flood vs. permanent water/land classifications) and wildfire scars (burned vs. unburned classifications). The metric used for evaluating these capabilities is Intersection over Union (IoU) for the subject category (Water/Burn Scar), as this is a much harsher form of assessing accuracy - as it enforces significant penalties against the model for producing false positives (e.g., mixing burn scar with created cloud shadow). Table 14 presents the impact of model scale and temporal context on disaster segmentation performance, highlighting how these factors influence IoU across different disaster scenarios.

TABLE 14. Impact of Model Scale and Temporal Context on Disaster Segmentation (IoU)

Model Configuration	Parameters	Flood IoU (Water)	Burn Scar IoU
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Prithvi-EO-2.0-TL (Time/Loc)	600M	83.1%	83.2%
Prithvi-EO-2.0 (No Metadata)	600M	82.5%	83.2%
Prithvi-EO-2.0	300M	82.2%	79.9%
Prithvi-EO-1.0 (Baseline)	100M	79.6%	76.8%

Scale is Important for Texture: Increasing the number of model parameters from 300M to 600M produced an increase in the Burn Scar IoU of 3.3%. However, the increase in Flood IoU was merely 0.9% [5]. Thus, it would appear that model scaling has a greater effect for detecting subtle spectral textures (such as ash contrasted with dark soil) than for detecting high-contrast items (such as water).

Temporal Context Decreases False Positives: For flood mapping tasks, in using the TL (time-location) configuration, it outperformed the standard on an IoU of 83.1 to 82.5. Therefore, that indicates that the model is better able to differentiate temporary from permanent water bodies when it accounts for seasonality versus when it is using only visual features.

Superior to Baseline Task-Specific Models: By way of example, in performing related dense tasks such as landslide detection, as listed in Table VII of the source, the foundation model produced a mean IoU of 71.3, while the baseline standard U-Net [18] produced a mean IoU of 70.4. Thus, this indicates that the large-scale pretraining of the foundation model offers a competitive advantage to identifying rare and complex hazards from small supervised datasets through its development of a "world model."

3. Temporal Change-Oriented Tasks

Task Relevance and Model Property:

Crop type mapping serves as the definitive test for temporal reasoning and phenological awareness in foundation models. In contrast to static objects (e.g., buildings), crops are dynamic

biological systems whose growth cycles unfold over time, requiring the model to assess whether. The model is treating the satellite data as a simple collection of "snapshots" or as a continuous, physically meaningful time series. And the model's notion of time is aligned with the physical world (e.g., crops have a specific schedule for growth).

Benchmarking Analysis:

To assess this capability, we will utilize the PASTIS [67] (Sentinel-2) time-series dataset to benchmark this capability against both 1) a collection of "General" Foundation Models and 2) a reference baseline (U-TAE, a U-Net with Temporal Attention [67]) that is specialized for this application. We will also conduct an experiment to assess performance in a few-shot (10% data) regime to assess data efficiency. As shown in Table 15, the performance of foundation models on time-series crop segmentation is strongly influenced by their ability to capture temporal dynamics and seasonal patterns, with temporally-aware architectures achieving higher mIoU.

TABLE 15. Comparative Performance on Time-Series Crop Segmentation (PASTIS) Metric: Mean Intersection over Union (mIoU). Higher is better.

Model Tier	Model Name	Architecture	mIoU (100% Data)	mIoU (10% Data)
SOTA FM	Prithvi-EO-2.0	ViT-Huge (3D)	53.4%	37.4%
Specialist	U-TAE (Baseline)	Temporal U-Net	52.9%	36.9%
Static FM	DOFA	ViT-Base (2D)	37.0%	20.4%
Static FM	Satlas	Swin-v2 (2D)	31.5%	15.8%
Pixel FM	Presto	Transformer (1D)	28.1%	20.1%

Benchmarking identifies three main architectural lessons:

Non-negotiable Temporal Modeling: Temporal context is more important than spatial resolution for vegetation tasks, as demonstrated by the large disparity between time-aware and static models ($>15\%$ mIoU). Flattening time or independently treating images does not provide adequate 'fingerprint' data (phenological) of the crop resulting in misclassification of crops during the massive pre-training phase.

Generalist FMs Outperform Specialist FMs: While specialist FMs (U-TAE) have historically outperformed generalist FMs in their ability to solve complex time-series problems, the transition to generalist FMs (Prithvi-EO-2.0 at 53.4%) outperforming specialist FMs (U-TAE at 52.9%) signals a new era for spatiotemporal vision transformers. A large, sufficiently spatiotemporal vision transformer will be able to create a universal time representation comparable to handcrafted temporal encoders.

Data Efficiency Dependent on Pre-training Objectives: In the 10% Data regime, while Prithvi maintains usable performance levels (37.4%), Satlas falls apart (15.8%). Satlas's reliance on supervised pre-training (label-specific) leaves it susceptible to lack of fine-tuning data, while MAE pre-training (learned physics of data) enables Prithvi to rapidly adjust to performing new tasks with limited supervision.

4. Small-Object & Geometry-Sensitive Tasks

Task Relevance and Model Property:

Local Inductive Bias and Multi-Scale Robustness of Foundation Models are challenged by small object detection (i.e., ships, vehicles) and geometry extraction (i.e., roads, building footprints). Unlike scene classification, where the "gist" is enough for classification, these tasks require the model to maintain high-frequency spatial detail. The ship size in a Sentinel-2 image is 10 m x 10 m; this is essentially one pixel in the higher (0.3 m) resolution and is visually identifiable. This task assesses the ability of Foundation Models to handle extreme scale variation and avoid small objects "vanishing" in the standard vision transformer patch token.

Benchmarking Analysis:

We will be using DOTA v1.0 [68] (Dataset for Object Detection in Aerial Images), the industry benchmark for oriented object detection and featuring thousands of small, rotated instances of ships, planes, and vehicles. For this study, we will compare Scale-MAE [23] (optimized for scale) to SatMAE [17] (standard temporal/spectral) and the standard ViT baselines. Table 16 presents a comparative evaluation of foundation models on the DOTA v1.0 benchmark, highlighting the challenges of detecting small and densely packed objects in high-resolution remote sensing imagery.

TABLE 16. Comparative Performance on Small Object Detection (DOTA v1.0) *Metric: mean Average Precision (mAP). Higher is better.*

Model	Backbone	Pre-training Strategy	mAP (DOTA)	Key Strength/Weakness
Scale-MAE	ViT-Large	Scale-Aware MAE	77.8%	SOTA. Explicit GSD encoding allows it to adapt features to object size (e.g., small ship vs large ship).
MAE (Standard)	ViT-Large	Masked Autoencoding	74.2%	Baseline. Struggles with scale variation; tends to miss extremely small objects that vanish in patches.
SatMAE	ViT-Large	Temporal MAE	73.5%	Underperforms. Optimized for <i>time</i> and <i>spectrum</i> (Sentinel-2), not high-res geometry (DOTA is aerial).

Supervised	ResNet-50	ImageNet Weights	72.1%	Lacks the domain-specific "texture" understanding of RS foundation models.
Satlas	Swin-v2	Multi-Task Supervised	<i>Mixed</i>	excels at "Marine Infrastructure" but lags on small vehicle counting due to label noise.

Three Important Architectural Learnings about DOTA Performance:

The "Vanishing Object" Problem associated with ViTs [7]: Standard Vision Transformers (like SatMAE [17]) create 16*16 pixel patch representations of images. When the size of an object is less than the patch size, or when it is straddled between 2 patches, its corresponding features will be averaged out during the self-attention process. Therefore, Scale-MAE [23] will perform better than SatMAE (+4.3 mAP) due to the fact that it employs a GSD-conditioned decoder, whereby the model is forced to reconstruct high-fidelity information (edge pixels of ships) at their actual scale vs. treating them as texture noise.

Variation in Image Acquisition Methods (Satellite vs. Aerial): The performance of the tested model (SatMAE) on this dataset (73.5%) is noticeably lower than its overall performance on the Segmentation test. This is a prime illustration of a domain gap: while SatMAE was pre-trained on Sentinel-2 imagery (10m resolution), DOTA consists of images taken from aerial vehicles (0.1m - 1m resolution). A model trained to identify "forest phenology" at a 10m resolution does not necessarily learn to identify "ship geometry" at a .5m resolution. This suggests that the resolution-specific pre-training of the model is generally more important than its size or scale with respect to geometry-related tasks.

The Lack of Explicit Geometry Encoding: Foundation models exceed ResNet's performance on the standard supervised baseline for classification (77.8% vs. 72.1%); however, this performance gain is much smaller when compared with the equivalent classification models for this test. The

reduced performance gain suggests that Masked Image Modeling (MIM) techniques, which are fundamentally based on the task of reconstructing pixel values, are not the most effective objectives for training geometry-based models. Future methodology (such as those embodied in RingMo [24]) should utilize explicit masking of "objects" rather than arbitrary "patches" to ensure that the algorithm will learn the structural relationship of elements in the image ("roads") rather than merely relying on the textures present within the image surface.

8. Challenges and Future Directions

Even though foundation models for remotely sensed images hold great potential for many types of downstream tasks, there are inherent problems with them due to the specially defined physical, spectral, and temporal characteristics of Earth observation data. This section summarizes these limitations and describes future research directions based on the behavioral analyses discussed in Section 6.

1. Representation Challenges in Remote Sensing FMs

Core Challenge: Inductive biases of standard Vision Transformers (ViTs) [7] that are used for natural images do not fully represent how things really are with respect to geophysical properties.

Spectral Discontinuity: As shown in hyperspectral analysis (Section 5.2.9 SpectralEarth), using standard patch tokenization produces a discontinuity in the spectral signature by treating a 16x16 patch as a layer that has been flattened out. Flattening a patch destroys the critical fine-scale spectral correlation needed for the identification of chemical composition or mineral type.

Texture Bias of MAE: With the Masked Autoencoding (MAE) [4] method, the input is treated only as an integer value. The method does not account for the physics of reflectance, sensor noise, and so on. As a consequence, most of the time atmospheric

artifacts (clouds and haze) will be reconstructed as part of the reconstruction process rather than accounted for in the identification of atmospheric artifacts.

Vanishing Geometry: As shown in Small Object benchmarks (Section 6.4), large structures, such as roads or rivers that occupy only small amounts of space in the patch, get 'blurred' due to global attention and may disappear altogether in the embedding space.

2. Pre-training Objective and Data Challenges

Main Challenge: Optimizing goals currently prioritizes the mathematical aspects (i.e., pixel support) rather than considering geophysical semantics.

Semantic v. Texture: The benchmark comparison results presented in Section 6.1 show that Satlas [48] (Weak Supervision) has the highest performance in semantic classification, and MAE [4] models have the highest performance in texture construction. Together, these demonstrate that using pixel replication alone is not sufficient for capturing the overall scene-level semantics.

Geographic Weak Supervision Bias: Weakly supervised methods are able to perform well, but they are using publicly available datasets (OpenStreetMap data); therefore, there will be a large difference in domain when trying to use these models in the Global South, where map labels may not be present or are incorrect.

Contrastive Learning Fragility: Contrastive learning (e.g., DINO [65]) requires very strong conditionings through the use of data augmentation (cropping, color variations); yet when using contrastive learning in the space of remote sensing, for example, color jittering can confuse the actual physical meaning of a pixel and produce a physically invalid representation (i.e., vegetation health index).

3. Architectural and Computational Challenges

Main Issues With Current Methods for Developing Remote Sensing Systems:

The only way to scale the size of your model is by using a larger model (e.g., an encoder with more than 600 million parameters); however, this does not solve operational remote sensing's unique operational remote sensing use-case constraints.

The Resolution-Scale Trade-off: As mentioned in the Scale-MAE [23] analysis (see Section 6.1), a "universal" encoder may be underperforming on fixed-conditions benchmark data. This is because there is significant architectural overhead associated with managing multiple resolutions, which can have a detrimental effect on performance with respect to standard tasks.

Quadratic Cost vs. Edge Deployment: Because of the $O(N^2)$ complexities involved with ViTs, ViTs [7] are computationally cost-ineffective for processing gigapixel satellite images or sending real-time disaster response based applications onboard satellites.

Small Object Detection: Pure ViTs have lower inductive biases associated with CNNs and, therefore, struggle with detecting small, oriented objects (i.e., ships and vehicles); in essence, their only solution is to introduce complex decoder assemblies that cause significant delays in the inference processing.

4. Evaluation, Robustness, and Deployment Challenges

Main Issue: Current benchmarks focus too heavily on the leaderboard metrics of F1/IoU by using only static datasets. This results in an overreliance on benchmark numbers to predict how site-specific the application will work in real life (out of the sample).

Fallacy of the "Static World": Current benchmarks like EuroSAT [69] and BigEarthNet [20] evaluate models on geographically fixed, temporally fixed, and seasonally fixed data without ever testing model generalization over domain shift (training on European forests

and testing on Amazonian rainforest) or sensor drift, the two main reasons for failure in the real world.

Blindness to Uncertainty: In disaster scenarios (see Section 6.2), a model predicting "flood" with a probability of 51% is weighted the same as several others predicting flood (e.g., 99%). The standard evaluation metric of mIoU does not reflect how well calibrated the model is (and consequently may misclassify data) and is, therefore, critical to assessing risk.

Out of Distribution for Disasters: Models are very seldom tested for ODD (Out of Distribution) with Black Swan events (e.g., 1 in 100-year service) that result in significant, visual differences from the model's originally trained dataset.

8. Conclusion

In this study we performed a narrow overview of foundation models utilized within remote sensing for image processing, focusing primarily on their design architectures for pre-training, objectives employed to impact downstream knowledge tasks of remote sensing images. In opposition to broader overviews discussed with respect to either Earth system modeling or generic forms of vision and language frameworks, our objective was to analyze foundation models functioning as omnipurpose image processors through the environmental-specific spectrospatial and temporal characteristics present in remote-sensing data.

We systematically organized the current models according to how they address each of these types of modal constraints, eliminating simply listing them sequentially and instead highlighting their inherent structural inductive bias.

A task-family-based analysis of scene-level classification, dense pixel-level prediction, and temporal regression highlighted a number of consistent behavioral patterns. Temporal context was by far the most important contributor to the semantic robustness of models; specifically, models that explicitly encoded temporal context performed better than strictly static architectures on vegetation-dominant tasks due to their superior ability to generalize across temporal environments. On the other hand, models that were optimized for universality (i.e.,

scale-invariant encoders) but whose architecture was constrained by defined methodologies often exhibited performance trade-offs when evaluated against fixed-condition benchmarks. This finding suggests that modeling architectures that allow for flexibility incur a cost to the model's discriminative ability. Models that were weakly supervised at pre-training (i.e., also used less data for building the backbone as compared to their fully supervised equivalents) were able to achieve similar performance levels to the fully supervised models, thereby making the importance of pre-training objectives nearly equal to that of model scaling.

Overall, the results of this report show that advancements will continue to be made in building out our ability to utilize realistic parametric representations of physical processes through investigating how those representations and model architectures are formed on the basis of creating a principled connection between physical measurements (data) and the fundamental assumptions (inductive biases) that any given model has. As the application of foundation models becomes more pervasive (to include sharing data across multiple users with different applications), it will become more challenging to evaluate them based solely on their ranking or performance relative to other models on a leaderboard. Therefore, we believe that we need to utilize more robust and rigorous methods to assess the quality of foundation models, particularly when it comes to their accuracy and strength of calibration, physical reality, and suitability for use on operationally viable remote sensing systems.

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